

Software Requirements Specification (SRS)

Project Title: Phobitron VR (Phobia Testing and Therapy using Virtual Reality)

1.Introduction:

1.1 Purpose:

Phobias are one of the most common mental health issues, affecting millions of individuals worldwide. They are characterized by an extreme, irrational fear of specific objects, situations, or activities, which can severely impact a person's daily life and well-being. Traditional exposure therapy involves gradually exposing patients to the feared object or situation, but this can be challenging due to logistical constraints or the risk of triggering severe anxiety. This paper outlines the software specifications for the Phobitron VR project, which intends to offer a virtual reality-based platform for phobia testing and treatment. This system will facilitate exposure therapy in a controlled environment, enabling users to confront their fears through immersive experiences.

The document aims to serve as a foundational guide for:

- **Software developers** who will implement the system features.
- **System designers and architects** responsible for overall design and integration.
- **Quality assurance teams** for defining test criteria.
- **Therapists and mental health professionals** who will utilize the application for treatment.
- **System administrators and stakeholders** involved in planning and deployment.

1.2 Scope of the System

The “Phobitron VR” project focuses on utilizing virtual reality technology to help individuals identify and treat their specific phobias in a safe and controlled virtual environment. The system is built to provide:

- **Immersive, customizable VR environments** tailored to multiple phobias.
- **Gradual exposure levels** managed in real-time by therapists.
- **Real-time monitoring** of patient vitals using biometric sensors and Gaze Tracker.
- **Automated report generation and data analytics** to assess progress.
- **Scheduling and notification system** for therapy sessions.
- **Gamification elements** to encourage participation and track achievements.
- **Remote and cross-platform access**, including VR headsets, tablets, and smartphones.
- **Support for multiple user roles**, including patients, therapists, and administrators.

The system is designed to be scalable, secure, and compliant with healthcare data regulations.

1.3 Definitions, Acronyms, and Abbreviations

Acronym	Description with Abbreviation
VR	Virtual Reality – A computer-generated simulation of a 3D environment that can be interacted with in a seemingly real way using special electronic equipment.
API	Application Programming Interface – A set of protocols and tools that allow different software applications to communicate with each other.
UI/UX	User Interface/User Experience – UI refers to how a user interacts with a system's interface, while UX is the overall experience and satisfaction of the user.
OTP	One-Time Password – A time-sensitive or single-use code used for user authentication during login.
SRS	Software Requirements Specification – A document that describes what the software will do and how it will be expected to perform.
AWS	Amazon Web Services – A cloud computing platform offering storage, processing, and networking services.
Phobia Module	A specific VR scenario or environment designed to simulate a particular phobia for exposure therapy purposes.

2. Overall Description

2.1 Product Perspective

“*Phobitron VR*” is an innovative software system that leverages Virtual Reality (VR) technology to treat phobias through immersive exposure therapy. It provides a safe, controlled and interactive environment where individuals can gradually confront and manage their specific fears. The system supports a wide range of phobias, including claustrophobia, arachnophobia and acrophobia.

By using VR, the project aims to overcome the limitations of traditional exposure therapy, enhancing both accessibility and therapeutic effectiveness. “*Phobitron VR*” is designed to be adaptable, scalable and engaging, offering a powerful tool for mental health professionals to improve patient outcomes

2.2 User Classes and Characteristics

- **Patients and People with Phobias**
 - The VR-based therapy will directly help end users deal with their particular anxieties.
 - Their input will influence the system's usability and efficacy.

- **Psychologists and therapists**
 - Core users that will oversee therapeutic sessions, track advancement, and modify situations.
 - Analytics, session management, and real-time physiological monitoring systems are necessary.
- **Doctors and Medical Facilities**
 - Phobitron VR is being used by organizations to provide mental health services.
 - Interested in scalability, system integration, and adherence to medical standards.
- **Investors**
 - Provide financial support for development, deployment, and scaling.
 - Focused on market potential, system viability, and return on investment.
- **Engineers and Developers**
 - Suppliers of VR headsets, biometric sensors, and other required devices.
 - Influence the system's hardware compatibility and performance.
- **Researchers**
 - Use data generated by the system for studies on phobia therapy and psychological research.
 - Require access to anonymized data and analytical insights.
- **Caregivers and Support Groups**
 - Assist patients during or after therapy sessions.
 - Interested in system reliability and effectiveness to support users.
- **Regulatory Authorities**
 - Ensure the system adheres to healthcare, data privacy, and security regulations.
 - Oversee compliance with industry standards.
- **Tech Enthusiasts and Innovators**
 - Explore the technological capabilities of VR for mental health applications.
 - Can contribute to future enhancements and partnerships.
- **Hardware Manufacturers**
 - Suppliers of VR headsets, biometric sensors, and other required devices.
 - Influence the system's hardware compatibility and performance.

2.3 Operating Environment

- **Hardware:** Oculus Rift, HTC Vive, biometric sensors, VR-compatible PC.
- **Software:**
 - VR Engine: Unity / Unreal Engine
 - Frontend: React.js
 - Backend: Python, Node.js
 - Database: PostgreSQL
 - Cloud: AWS or Azure

- **OS Support:**
 - Windows 10/11 for VR
 - Cross-platform (Web/Mobile) for dashboards

2.4 Design and Implementation Constraints

- VR latency must be under 20 milliseconds to prevent nausea.
- Compatible only with supported VR devices.
- Support for real-time multi-user sessions.
- Interfaces must be accessible (voice commands, subtitles, etc.).
- Codebase must support modular updates (e.g., adding new phobia modules).

3.Functional Requirements

Functional requirements are those that serve as examples for the system's internal operation. It comprises the task that the system should complete, the associated processes, the data that the system should store, and the user interfaces. These are the requirements that the end user specifically demands as basic facilities that the system should offer.

A User Story is a short, simple description of a software feature from the perspective of the end user. It is commonly used in **Agile** and **Scrum Methodologies** to capture user needs in a concise and understandable way.

Functional Requirements of Phobitron with User Story

FR-1: Customizable VR Therapy Environment

As a patient, I want to experience immersive and realistic VR environments so that I can confront my specific phobia safely and gradually.

FR-2: Gradual Exposure Levels

As a therapist, I want to guide patients through gradual levels of exposure so that therapy can be done safely without overwhelming them.

FR-3: Real-Time Physiological Monitoring

As a therapist, I want to monitor the patient's physiological responses in real-time so that I can assess stress levels and intervene when necessary.

FR-4: Automated Therapy Report Generation

As a therapist, I want the system to generate a therapy report automatically after each session so that I can review and adjust future therapy plans.

FR-5: User Progress Dashboard

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As a patient, I want to view my therapy progress on a dashboard so that I can stay motivated and informed about my improvement.

FR-6: Predictive Analytics

As a therapist, I want to receive AI-based therapy suggestions so that I can personalize treatment based on past session data.

FR-7: Therapist Scenario Control

As a therapist, I want to control and adjust the exposure scenario in real-time so that I can respond to the patient's anxiety levels effectively.

FR-8: Multi-Phobia Support

As a patient, I want to choose from a variety of phobia modules so that I can receive therapy tailored to my specific fear.

FR-9: Gamification Elements

As a patient, I want to earn badges and rewards during therapy so that I feel more motivated and encouraged to continue my sessions.

FR-10: Session Scheduling and Notifications

As a patient, I want to schedule therapy sessions and receive reminders so that I don't miss my appointments.

As a therapist, I want to manage my session calendar so that I can allocate time for each patient efficiently.

FR-11: Multi-User Roles

As an admin, I want to define different access roles for patients, therapists, and admins so that sensitive information is properly managed.

FR-12: Secure Login & Identity Verification

As a user, I want to log in securely using email, password, and OTP so that my personal therapy data is protected from unauthorized access.

FR-13: Remote Therapy Support

As a patient, I want to participate in therapy sessions from home so that I can receive treatment without traveling to a clinic.

FR-14: Notification System

As a patient, I want to get notified about upcoming sessions and progress updates so that I stay on track with my therapy.

As a therapist, I want to send important alerts to patients to ensure better communication.

FR-15: Emergency Stop Button

As a patient, I want an emergency stop option during therapy so that I can immediately exit the session if I feel overwhelmed or anxious.

4.Non-Functional Requirements

Non-functional requirements are specifications that define the **system's operational characteristics**, such as **performance, scalability, availability, and maintainability**.

Non-Functional Requirements of Phobitron

NFR-1:Performance

VR scenes must render with **low latency** to avoid nausea/discomfort

NFR-2:Security

All data encrypted, shared only with therapist

NFR-3:Usability

UI should be simple for patients, advanced options for therapists

NFR-4:Reliability

System should be fault-tolerant and operate without crashes during therapy sessions

NFR-5:Availability

The system shall maintain an availability of at least **99.5% uptime**, excluding planned maintenance.

NFR-6:Compliance

Adhere to medical privacy laws.

NFR-7:Accessibility Support

Voice navigation, adjustable font size, subtitles, colorblind modes.

NFR-8:Compatibility

Must work across major VR platforms (HTC Vive, PC, mobile/tablet for tracking).

NFR-9:Scability

Support concurrent sessions for multiple users and therapists.

NFR-10: Maintainability

Codebase should allow for easy updates (e.g., adding new phobia modules).

5. Requirement Prioritization

Prioritize the requirement for “Phobitron” according to the MoSCoW, Kano Model, or 100-Point Method.

MoSCoW Model

MoSCoW stands for:

Must-Have: Critical requirements that are essential for the system to function.

Should-Have: Important but not critical requirements; can be included in a later release.

Could-Have: Nice-to-have features that add value but are not essential.

Won't-Have (for now): Low-priority requirements that can be postponed.

Requirement	MoSCoW Category	Reason
Customizable VR scenes	Must-Have	Core mechanic of exposure therapy
Gradual exposure levels	Must-Have	Ensures patient safety during desensitization
Real-time physiological monitoring	Must-Have	Allows therapist/AI to gauge anxiety instantly
Automated therapy reports	Must-Have	Required for clinical documentation & feedback
User progress dashboard	Should-Have	Supports motivation & therapist review but not critical for first run
Predictive analytics (AI suggestions)	Could-Have	Valuable add-on once baseline therapy works
Therapist live scene control	Must-Have	Clinically essential to pause/intensify exposure

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Multi-phobia content library	Must-Have	Broad treatment coverage expected by users
Gamification badges	Could-Have	Nice-to-have engagement booster
Session scheduling & reminders	Must-Have	Prevents no-shows; integral to therapy workflow
Role-based access (patients / doctors / admin)	Must-Have	Security & data-segregation baseline
Secure login & OTP	Must-Have	Protects sensitive medical data
Remote (self-guided) therapy mode	Should-Have	Expands reach; not mandatory for MVP
Notification center	Should-Have	Improves engagement but therapy can run without it
Emergency stop button	Must-Have	Patient safety requirement
Low-latency VR performance	Must-Have	Prevents nausea and therapy failure
Data encryption & privacy	Must-Have	Legal/ethical compliance (HIPAA/GDPR)
Simple + advanced usability modes	Must-Have	Patients need simplicity; doctors need control
Reliability	Must-Have	Therapy must not crash mid-session
Availability	Must-Have	Users may schedule outside office hours
Regulatory compliance docs	Must-Have	Mandatory for any medical-class software
Accessibility (voice, fonts, subtitles)	Should-Have	Inclusive design; can follow core launch
Multi-platform compatibility	Must-Have	Oculus, Vive, PC, mobile tracking – widens access
Scalability	Should-Have	Needed when user base grows
Maintainability	Should-Have	Eases future phobia-module updates

Kano Model

The Kano Model classifies requirements into five categories based on user satisfaction:

Must-Be (Basic Needs): Users expect these; missing them causes dissatisfaction.

Performance (Linear Features): More is better.

Excitement (Delighters): Unexpected features that surprise and delight users.

Indifferent: Features that do not significantly affect user experience.

Requirement	Kano Category	Explanation
Secure login & OTP	Must-Be	Absence causes instant distrust

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Role-based access	Must-Be	Basic security expectation
Data encryption & privacy	Must-Be	Mandatory compliance
Emergency stop button	Must-Be	Safety baseline
Multi-phobia support	Must-Be	Users expect variety of phobias
Gradual exposure levels	Performance	Higher satisfaction
Customizable VR scenes	Performance	Better outcome
Session scheduling & reminders	Performance	Convenient scheduling improves experience
Low-latency rendering	Performance	Less discomfort
Availability	Performance	Happier users
Reliability	Performance	Better trust
Multi-platform compatibility	Performance	Broader value
Real-time physiological monitoring	Excitement	Unexpected tech that delights users & doctors
Automated therapy reports	Excitement	Saves clinicians' paperwork; surprise benefit
Predictive analytics	Excitement	AI guidance exceeds baseline expectation
Gamification badges	Excitement	Pleasant surprise
Accessibility options	Excitement	Delights users with special needs
Remote self-guided mode	Performance	Convenience scales with availability
Notification center	Basic	Modern apps are expected to notify
Simple + advanced UI modes	Basic	Usability expectation
Maintainability	Indifferent	End-users rarely notice but developer care
Scalability	Indifferent	Invisible until heavy load
Compliance docs	Indifferent	Users assume it's handled
Sensor gateway (tech detail)	Indifferent	End-users unaware of backend

100 Points Model

100 points model work in this way

Each stakeholder gets 100 points to distribute among different requirements based on their importance.

The higher the points assigned, the more critical the feature.

The final scores help in prioritization.

Requirement	Points	Short Rationale
Customizable VR scenes	10	Core therapy element
Real-time physiological monitoring	9	Critical feedback loop

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Gradual exposure levels	8	Directly tied to treatment safety
Therapist live control	7	Mandatory in clinician-led sessions
Session scheduling & reminders	6	Prevents missed therapy
Automated therapy reports	6	Saves clinician time, regulatory need
Low-latency VR performance	5	Key QoE driver
Secure login & OTP	4	Baseline security
Data encryption & privacy	4	Compliance risk mitigator
User progress dashboard	4	Motivation / insight
Multi-phobia support	4	Breadth of service
Availability	3	User trust & uptime
Reliability / fault tolerance	3	Avoids session disruption
Predictive analytics	3	Future-value AI
Role-based access	3	Segregates permissions
Multi-platform compatibility	3	Wider adoption
Remote therapy support	3	Convenience & reach
Notification Center	3	Engagement booster
Emergency stop	4	Safety gate
Accessibility options	2	Inclusivity
Gamification badges	2	Motivation, lower impact
Maintainability	2	Long-term cost control
Scalability	2	Future-proofing
Compliance documentation	2	Audit requirement
Sensor gateway tech	1	Backend concern only

6. Validation and Verification Plan

6.1 Validation Techniques

Validation ensures that the Phobitron system meets user needs and expectations. The following techniques will be used:

- **Walkthroughs:** Conducted with therapists and patients to review prototypes and mockups, ensuring the system aligns with real-world workflows.
- **User Feedback:** Collecting direct feedback through surveys and usability testing sessions during prototype evaluation.
- **Reviews:** Regular reviews of requirements and design artifacts with stakeholders to confirm completeness and correctness.

These validation activities help confirm that the final product fulfills the intended user stories and functional requirements.

6.2 Verification Techniques

Verification ensures that Phobitron has been built correctly according to the documented requirements and specifications. The following techniques will be applied:

- **Inspections:** Structured examination of SRS, design documents, and prototypes to detect inconsistencies and ambiguities.
- **Testing:** Functional testing of each feature (FR-1 to FR-15) to ensure compliance with specified requirements.
- **Traceability:** Maintaining bidirectional traceability between requirements, design, and test cases to confirm coverage.

These techniques ensure that each requirement is implemented correctly and functions as expected.

6.3 Requirement Traceability Matrix (RTM)

The Requirement Traceability Matrix links functional requirements to their corresponding test cases to ensure complete test coverage.

Requirement ID	Requirement Title	Test Case IDs
FR-1	Customizable VR Therapy Environment	TC-1, TC-2
FR-2	Gradual Exposure Levels	TC-3
FR-3	Real-Time Physiological Monitoring	TC-4
FR-4	Automated Therapy Report Generation	TC-5
FR-5	User Progress Dashboard	TC-6
FR-6	Predictive Analytics	TC-7
FR-7	Therapist Scenario Control	TC-8

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Requirement ID	Requirement Title	Test Case IDs
FR-8	Multi-Phobia Support	TC-9
FR-9	Gamification Elements	TC-10
FR-10	Session Scheduling and Notifications	TC-11, TC-12
FR-11	Multi-User Roles	TC-13
FR-12	Secure Login & Identity Verification	TC-14
FR-13	Remote Therapy Support	TC-15
FR-14	Notification System	TC-16, TC-17
FR-15	Emergency Stop Button	TC-18
NFR-1	Performance (VR scenes render with low latency)	QC-1
NFR-2	Security (data encryption & sharing control)	QC-2
NFR-3	Usability (simple UI for patients; advanced options for therapists)	QC-3
NFR-4	Reliability (fault-tolerant, no crashes)	QC-4
NFR-5	Availability ($\geq 99.5\%$ uptime)	QC-5
NFR-6	Compliance (with medical privacy laws)	QC-6
NFR-7	Accessibility Support (voice nav, subtitles, colorblind modes)	QC-7
NFR-8	Compatibility (across VR platforms)	QC-8

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NFR-9	Scalability (support concurrent sessions)	QC-9
NFR-10	Maintainability (easy updates, add modules)	QC-10

7. UML Diagrams

Use Case:

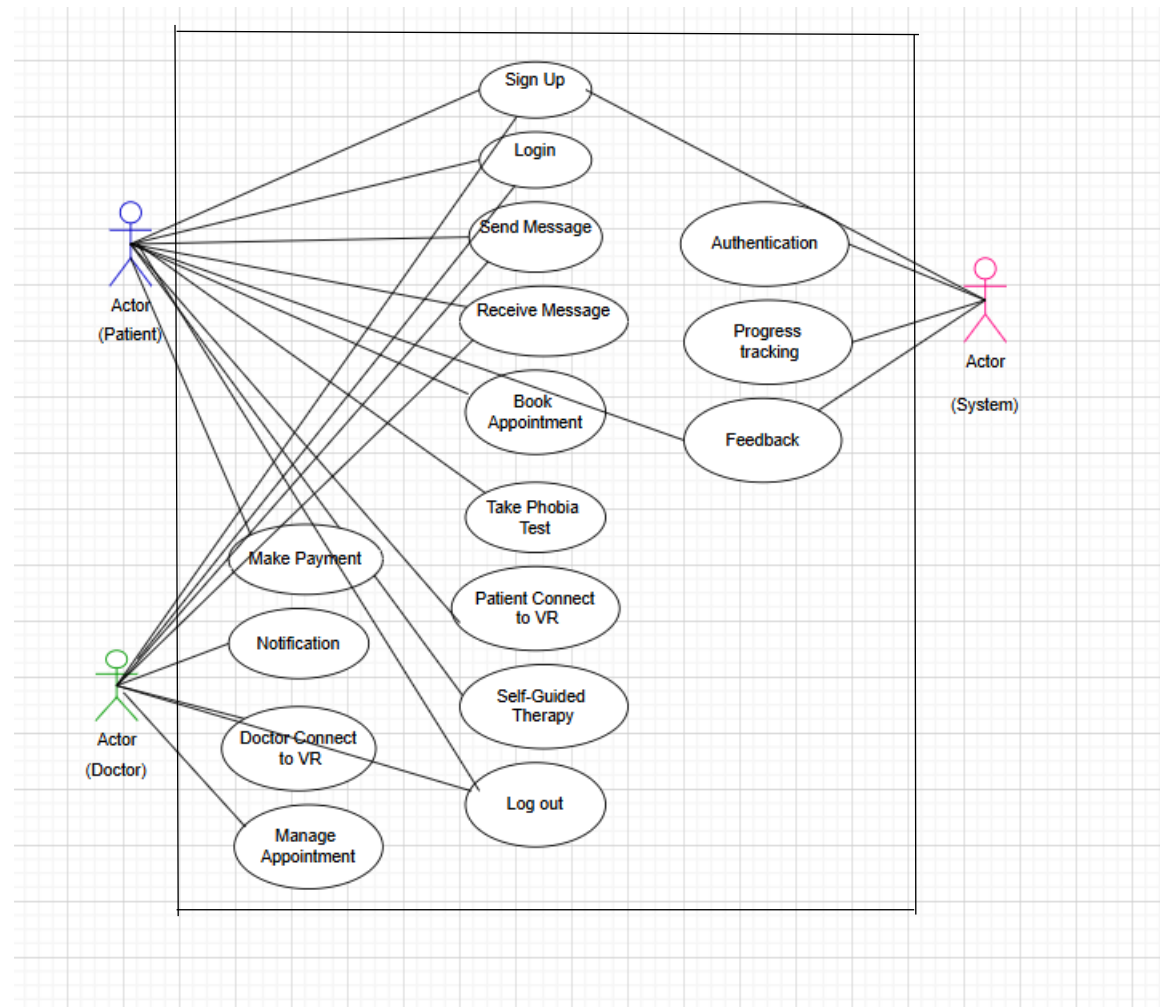


Figure 7.1: Use Case Diagram for Phobitron

Use Case Description:

Table 7.1: Sign Up

Use Case No.	01
Use Case Name	Sign Up
Goal	Register a user into the system.
Preconditions	The user is not already registered.
Success End Condition	The system successfully registers the user and creates an account.
Failed End Condition	The registration process fails, and no account is created.
Primary Actors:	Patient, Doctor

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Secondary Actors:	System	
Trigger	The user clicks the “Sign Up” button.	
Main Success Flows	Step	Action
	1	The user accesses the registration page.
	2	The system displays the registration form.
	3	The user fills out required details such as username, email, password, phone no, date of birth and user type (patient or doctor).
	4	The user submits the registration form.
	5	The system validates the submitted information.
	6	If valid, the system creates a new user account.
	7	The system notifies the user of successful registration.
Alternative Flows	Step	Action
	1	If the information is incomplete or incorrect, the system prompts the user to fix errors.
	2	The user corrects and resubmits the form.
	3	The system re-validates and proceeds if the data is now correct.
Quality Requirements	Step	Requirement
	1	Registration data must be securely transmitted and stored.
	2	Process should be quick and intuitive for all user roles.
	3	The system should detect duplicate accounts and prevent re-registration.
	4	The interface must work across various devices and browsers.

Table 7.2: Log In

Use Case No.	02
Use Case Name	Log In
Goal	The goal of the "Login" use case is to authenticate and authorize users, providing them access to their individual accounts within the Phobitron System. This feature ensures the security of user accounts and allows patients and medical professionals to access personalized information and functionalities.
Preconditions	The user is registered in the Phobitron System and The user has a valid username and password.
Success End Condition	The user successfully logs into the Phobitron System and gains access to their account.
Failed End Condition	The login attempt fails due to incorrect credentials, account suspension, or technical issues.
Primary Actors:	Patients, Medical professionals

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Secondary Actors:	System	
Trigger	The user initiates the login process by entering their username and password.	
Main Success Flows	Step	Action
	1	The user accesses the login page.
	2	The system displays the login form.
	3	The user enters their username and password
	4	The user submits the form.
	5	The system verifies the entered credentials.
	6	If valid, the system redirects the user to their dashboard and grants access.
Alternative Flows	Step	Action
	1	If credentials are incorrect or incomplete, the system shows an error message.
	2	The user re-enters the correct information.
	3	The system re-verifies and proceeds with login if valid.
Quality Requirements	Step	Requirement
	1	Login credentials must be encrypted during transmission and storage.
	2	Login should complete within 3 seconds under normal conditions.
	3	System must implement account lockout after multiple failed login attempts to prevent brute-force attacks.
	4	System should remain responsive under high login traffic without performance degradation.

Table 7.3: Send Message

Use Case No.	03	
Use Case Name	Send Message	
Goal	Enable the patient or doctor to send a message within the system.	
Preconditions	Both sender and recipient must have active accounts.	
Success End Condition	Message is successfully delivered to the intended recipient.	
Failed End Condition	Message fails to send due to technical issues.	
Primary Actors:	Patient, Doctor	
Secondary Actors:	System	
Trigger	User clicks the “Send” icon after composing a message.	
Main Success Flows	Step	Action
	1	User navigates to the messaging interface.
	2	User selects a recipient.

	3	User composes a message.
	4	User clicks the send button.
	5	System saves and delivers the message to the recipient's inbox.
Alternative Flows	Step	Action
	1	If the recipient is not available, the system queues the message.
	2	If sending fails, system notifies the sender and saves to drafts.
Quality Requirements	Step	Requirement
	1	Messages must be securely transmitted using encryption.
	2	User should be able to view message history.
	3	Message delivery must be confirmed in real-time.

Table 7.4: Receive Message

Use Case No.	04	
Use Case Name	Receive Message	
Goal	Allow users to receive communication from others within the platform.	
Preconditions	The sender must have sent a message. The recipient must be logged in.	
Success End Condition	Message is successfully delivered and viewed by the recipient.	
Failed End Condition	Message is not received or displayed due to a system issue.	
Primary Actors:	Patient, Doctor	
Secondary Actors:	System	
Trigger	System detects a new incoming message for the logged-in user.	
Main Success Flows	Step	Action
	1	Message is sent by another user.
	2	System notifies the recipient of a new message.
	3	Recipient opens the messaging interface.
	4	System displays the message content.
	5	Recipient can reply the message.
Alternative Flows	Step	Action
	1	If the user is offline, system stores and notifies upon next login.
	2	If the message fails to load, system prompts to retry.
Quality Requirements	Step	Requirement
	1	Notifications must be real-time and reliable.
	2	Message storage should be persistent and secure.

	3	User must be able to distinguish new/unread messages.
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Table 7.5: Book Appointment

Use Case No.	05	
Use Case Name	Book Appointment	
Goal	Enable a patient to schedule an appointment with a doctor.	
Preconditions	The patient must be logged in and have access to available doctors.	
Success End Condition	The appointment is successfully booked and saved in the system.	
Failed End Condition	Booking fails and the appointment is not saved.	
Primary Actors:	Patient	
Secondary Actors:	System, Doctor	
Trigger	The patient selects a doctor and clicks the “Book” button	
Main Success Flows	Step	Action
	1	The patient navigates to the appointment booking page.
	2	The system displays a list of available doctors and time slots.
	3	The patient selects a doctor and time slot.
	4	The patient confirms and submits the appointment request.
	5	The system saves the appointment and sends notifications to both parties.
	6	The appointment is shown in the patient and doctor dashboards.
Alternative Flows	Step	Action
	1	If no time slots are available, the system shows a message to try a different time or doctor.
	2	If saving fails due to a system error, the user is prompted to retry.
Quality Requirements	Step	Requirement
	1	Appointment slots must update in real time to avoid double-booking.
	2	Notifications must be instant and reliable.
	3	Booking system must be responsive and load within 2 seconds.
	4	Users should receive confirmation via email, message or in-app alerts.

Table 7.6: Take Phobia Test

Use Case No.	06	
Use Case Name	Take Phobia Test	
Goal	Allow the patient to take a psychological test to identify phobia types.	
Preconditions	The patient is logged in and has access to the test section and must have VR headset.	
Success End Condition	The patient completes the test and receives results.	
Failed End Condition	The test is incomplete or errors occur in submitting answers.	
Primary Actors:	Patient	
Secondary Actors:	System	
Trigger	The patient clicks on “Take a Phobia Test” button.	
Main Success Flows	Step	Action
	1	Patient opens the dashboard page and clicks on “Take a Phobia Test”
	2	Select a phobia which user would like to test and then clicks on “Start Test” button.
	3	System loads test instructions and questions.
	4	Patient completes the test by answering all questions.
	5	System processes and evaluates the answers.
	6	Test results are displayed and saved to the patient profile.
Alternative Flows	Step	Action
	1	Patient leaves some questions unanswered.
	2	System prompts the patient to complete all questions before submission.
Quality Requirements	Step	Requirement
	1	Questions and logic must be based on validated clinical standards
	2	Results must be stored securely and accurately.
	3	Test interface must be mobile-friendly and responsive.

Table 7.7: Patient Connect to VR Room

Use Case No.	07
Use Case Name	Patient Connect to VR Room
Goal	Allow the patient to connect to the assigned VR therapy session.
Preconditions	Patient has a compatible VR device and is scheduled for a session.

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Success End Condition	Patient connects successfully to the VR room.	
Failed End Condition	Patient cannot access the VR session.	
Primary Actors:	Patient	
Secondary Actors:	System	
Trigger	Patient starts VR session from the Appointment page by clicking on “Join” button.	
Main Success Flows	Step	Action
	1	Patient checks the schedule for VR session from Appointment Page and then join the session by clicking on Join button.
	2	System prompts for VR connection.
	3	Patient connects and confirms readiness.
	4	System validates credentials and enters patient into the VR room.
	5	Therapy session begins.
Alternative Flows	Step	Action
	1	VR device not detected – system shows help instructions.
	2	Access validation fails – system denies entry and logs error.
Quality Requirements	Step	Requirement
	1	Low latency streaming to prevent VR sickness.
	2	Device compatibility and connection diagnostics required.
	3	Session data must sync with patient progress logs.

Table 7.8: Self-Guided Therapy

Use Case No.	08	
Use Case Name	Self-Guided Therapy	
Goal	Allow patients to begin and complete light therapy sessions independently and find their phobia type and level so that they can make better decision and consult a specialist for their further therapy to recover.	
Preconditions	Patient is logged in and assigned self-guided therapy modules.	
Success End Condition	Patient completes the session and data is saved.	
Failed End Condition	Session fails to start or data is lost.	
Primary Actors:	Patient	
Secondary Actors:	System	
Trigger	Patient selects Take a Phobia Test module from dashboard.	
Main Success Flows	Step	Action
	1	Patient navigates to Take a Phobia Test page.

	2	System displays list of available phobia therapy modules.
	3	Patient selects a session and starts it.
	4	System loads the session content (text, audio, video, exercises).
	5	Patient interacts with the session content if he/she feels uncomfortable ,he/she can stop the session immediately.
	6	Upon completion, system saves session data to progress records.
Alternative Flows	Step	Action
	1	If content fails to load, system provides retry option.
	2	If session is incomplete, system prompts for continuation or exit.
Quality Requirements	Step	Requirement
	1	Session must be interactive and user-friendly.
	2	Data must sync to progress log securely.

Table 7.9: Make Payment

Use Case No.	09	
Use Case Name	Make Payment	
Goal	Enable the patient to complete financial transactions for services.	
Preconditions	User is logged in and has appointments, sessions, etc. that require payment.	
Success End Condition	Payment is successfully processed and recorded.	
Failed End Condition	Payment fails due to system error or invalid credentials.	
Primary Actors:	Patient	
Secondary Actors:	System, Payment Gateway	
Trigger	User selects a service and clicks the "Make due payment" button.	
Main Success Flows	Step	Action
	1	Patient request for an appointment or a therapy then clicks on "Make due payment" button.
	2	System displays the list of items and total amount due.
	3	User selects a payment method (e.g., card, mobile banking).
	4	User enters payment details and confirms.
	5	System sends the transaction to the payment gateway.
	6	On success, system updates payment status and notifies both the patient and doctor.

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Alternative Flows	Step	Action
	1	If payment is declined or errors occur, system notifies the user.
	2	User is prompted to retry or use a different payment method.
Quality Requirements	Step	Requirement
	1	Transactions should complete within 5 seconds on average.
	2	Email/SMS/in-app receipt should be auto-generated upon success.
	3	System must handle payment failures gracefully and provide retry options.

Table 7.10: Notification

Use Case No.	10	
Use Case Name	Notification	
Goal	Alert users about important events such as appointments, messages or updates.	
Preconditions	User is logged in the Phobitron.	
Success End Condition	Notification is delivered and viewed by the user.	
Failed End Condition	Notification is not delivered.	
Primary Actors:	Patient, Doctor	
Secondary Actors:	System	
Trigger	An event (e.g., new appointment, message, make payment) occurs.	
Main Success Flows	Step	Action
	1	System detects a new event (e.g., appointment scheduled).
	2	System creates a notification entry.
	3	Notification appears on user interface or is sent via email/SMS.
	4	User opens and views the notification.
Alternative Flows	Step	Action
	1	If notification fails, system retries.
	2	If user does not view, reminder is sent later.
Quality Requirements	Step	Requirement
	1	Notifications must be delivered in real-time.
	2	Should support multi-channel delivery (email, in-app, SMS).
	3	Users should be able to customize notification preferences.

Table 7.11: Doctor Connect to VR Room

Use Case No.	11	
Use Case Name	Doctor Connect to VR Room	
Goal	Enable the doctor to join a scheduled VR therapy session to guide the patient.	
Preconditions	Doctor has a compatible VR device and an upcoming session with a patient.	
Success End Condition	Doctor joins the VR room and is able to interact with the patient.	
Failed End Condition	Doctor fails to connect or cannot guide the session.	
Primary Actors:	Doctor	
Secondary Actors:	System	
Trigger	Doctor selects the “Create Room” or “Schedule For Later” button then join the session from patient history page.	
Main Success Flows	Step	Action
	1	Doctor logs in and checks the appointments for today section.
	2	By clicking on the patient name, there will appear patient history. Doctor will select phobia and therapy types and create VR room .
	3	System prompts to connect to VR headset.
	4	Doctor connects and confirms readiness.
	5	System authenticates session and grants access.
	6	Doctor is placed into the shared VR room with the patient.
Alternative Flows	Step	Action
	1	System fails to detect headset – troubleshooting tips displayed.
	2	Authentication fails.
Quality Requirements	Step	Requirement
	1	Doctor must have moderator access in the VR session.
	2	Communication between patient and doctor should be real-time and secure.
	3	VR data sync must be seamless and protected.

Table 7.12: Manage Appointment

Use Case No.	12	
Use Case Name	Manage Appointment	
Goal	Allow doctors to view, approve, decline or reschedule patient appointments.	
Preconditions	Doctor is logged in and has assigned or requested appointments.	
Success End Condition	Appointment status is updated or confirmed.	
Failed End Condition	Changes are not saved or operation fails.	
Primary Actors:	Doctor	
Secondary Actors:	Patient, System	
Trigger	Doctor accesses the appointments section.	
Main Success Flows	Step	Action
	1	Doctor logs into the dashboard.
	2	Doctor navigates to the appointment section.
	3	System displays list of requested and scheduled appointments.
	4	Doctor reviews, reschedules, or cancels appointments.
	5	System saves updates and notifies the patient.
Alternative Flows	Step	Action
	1	If the appointment cannot be changed (e.g., past date), system notifies the doctor.
	2	If update fails, system prompts retry.
Quality Requirements	Step	Requirement
	1	Appointment changes must be recorded in logs.
	2	Patient must be notified of doctor-initiated changes.
	3	System must prevent overlapping schedules.

Table 7.13: Authentication

Use Case No.	13	
Use Case Name	Authentication	
Goal	Allow both doctors and patients to securely log into the platform and access relevant features.	
Preconditions	User (doctor or patient) has an active account with valid credentials.	
Success End Condition	User is authenticated and redirected to their respective dashboard.	
Failed End Condition	Authentication fails due to incorrect credentials or technical errors.	
Primary Actors:	Doctor, Patient	
Secondary Actors:	System	
Trigger	User attempts to log in	
Main Success Flows	Step	Action
	1	User enters their login credentials.
	2	System validates the credentials.
	3	If valid, user is redirected to their appropriate dashboard.
Alternative Flows	Step	Action
	1	If credentials are incorrect, system displays an error.
	2	On repeated failure, system locks the account temporarily.
Quality Requirements	Step	Requirement
	1	Must support OTP verification for enhanced security.
	2	Credentials must be encrypted during transmission and storage.
	3	Interface must be user-friendly with password recovery option.

Table 7.14: Progress Tracking

Use Case No.	14
Use Case Name	Progress Tracking
Goal	Allow the patient or doctor to view progress data over time.
Preconditions	The user must be logged in and have previous therapy sessions recorded.
Success End Condition	The progress report is displayed accurately and completely.
Failed End Condition	Progress data cannot be fetched or displayed.
Primary Actors:	Patient, Doctor
Secondary Actors:	System

Trigger	Patient clicks on "Progress" or Doctor clicks on "Report" from their respective dashboard.	
Main Success Flows	Step	Action
	1	Patient navigates to progress tracking page from his dashboard. Doctor navigates to progress tracking page from patient history.
	2	System fetches past session data.
	3	System generates visual or tabular reports.
	4	Patient or Doctor can view or download the report.
Alternative Flows	Step	Action
	1	If data fetch fails, system displays a retry option.
	2	If no data exists, system informs the user.
Quality Requirements	Step	Requirement
	1	Reports must be accurate and updated in real-time.
	2	Visualization should be clean and understandable.
	3	Progress data should be secured and tamper-proof.

Table 7.15: Feedback

Use Case No.	15	
Use Case Name	Feedback	
Goal	Enable patients to submit feedback on their therapy experience or the platform.	
Preconditions	Patient is logged in and has interacted with therapy or system features.	
Success End Condition	Feedback is submitted and stored securely.	
Failed End Condition	Feedback submission fails due to error or incomplete input.	
Primary Actors:	Patient	
Secondary Actors:	System	
Trigger	Patient accesses feedback form from Appointment page.	
Main Success Flows	Step	Action
	1	Patient opens the feedback form.
	2	Patient fills out experience comments or rating.
	3	Patient submits the form.
	4	System confirms submission and stores the data.
Alternative Flows	Step	Action
	1	If form is incomplete, system alerts the user.
	2	If submission fails, patient is asked to retry.
Quality Requirements	Step	Requirement
	1	Feedback form must be easy to use and mobile-friendly.

	2	Confirmation message must be shown after submission.
	3	Feedback should be reviewed by platform admins.

Table 7.16: Log Out

Use Case No.	16	
Use Case Name	Log Out	
Goal	Allow the user to securely exit the system.	
Preconditions	User is logged in.	
Success End Condition	User is logged out and redirected to the login page.	
Failed End Condition	Logout fails and session remains active.	
Primary Actors:	Patient, Doctor	
Secondary Actors:	System	
Trigger	User clicks on the “Logout” button.	
Main Success Flows	Step	Action
	1	User clicks logout.
	2	System ends the session and clears session data.
	3	System redirects user to login or home page.
Alternative Flows	Step	Action
	1	If logout fails, user is shown an error and offered retry.
Quality Requirements	Step	Requirement
	1	Session termination must occur within 1 second.
	2	User should receive confirmation of successful logout.
	3	Session timeout should also trigger auto-logout.

Class Diagram:

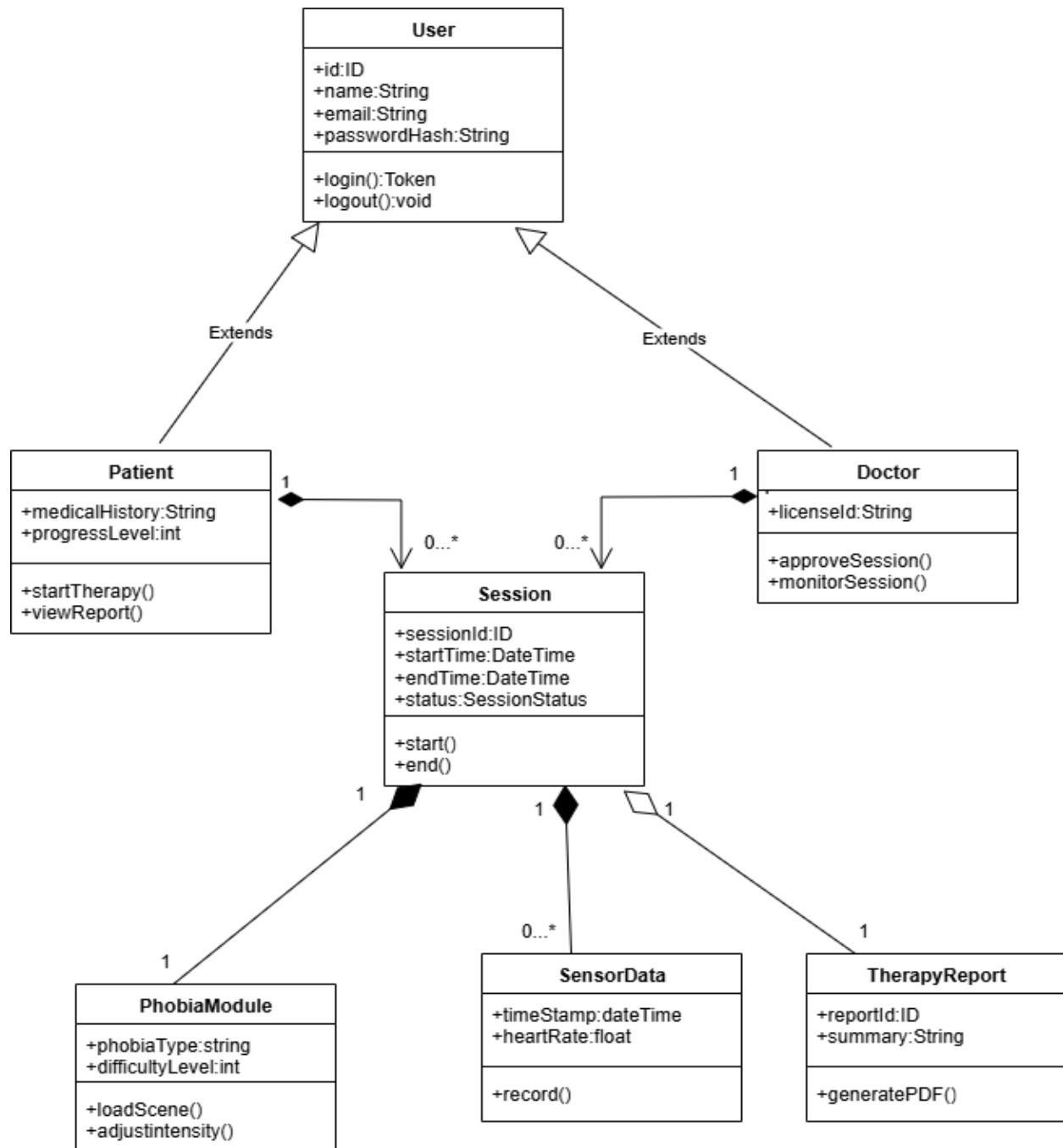


Figure 7.2: Class Diagram of Phobitron's Domain Entity

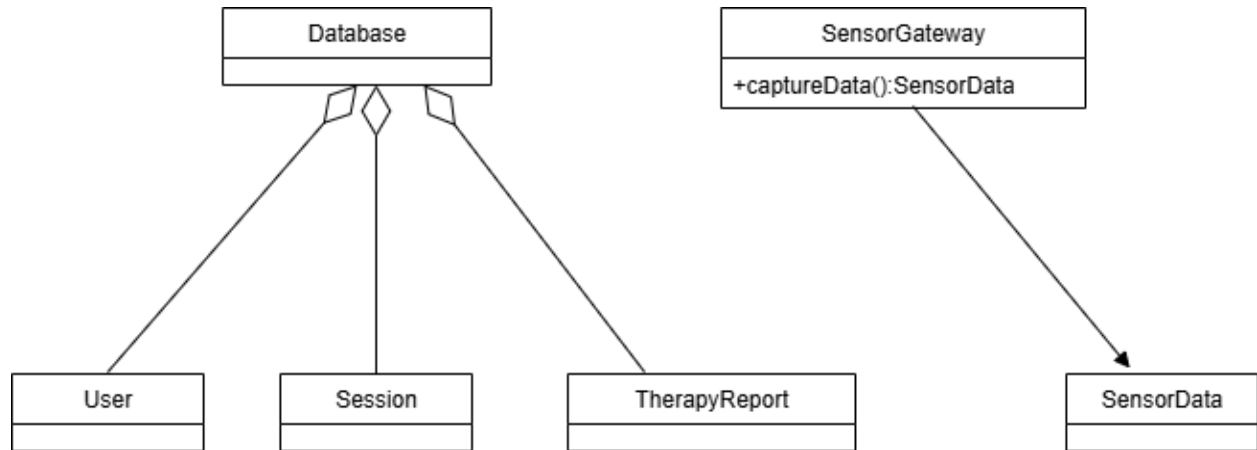


Figure 7.3: Class Diagram of Phobitron's Infrastructure

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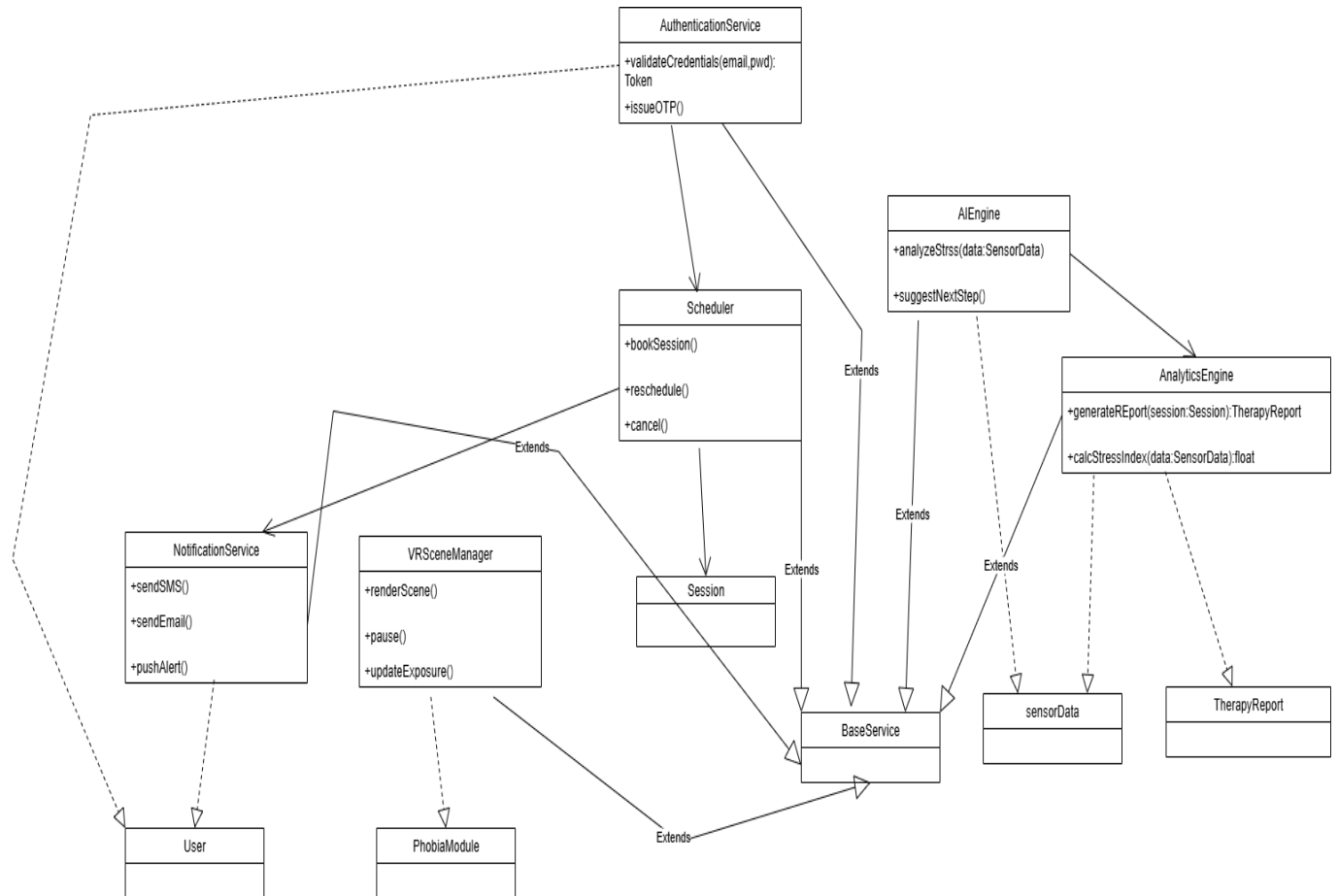


Figure 7.4: Class Diagram of Phobitron's Services

Sequence Diagram:

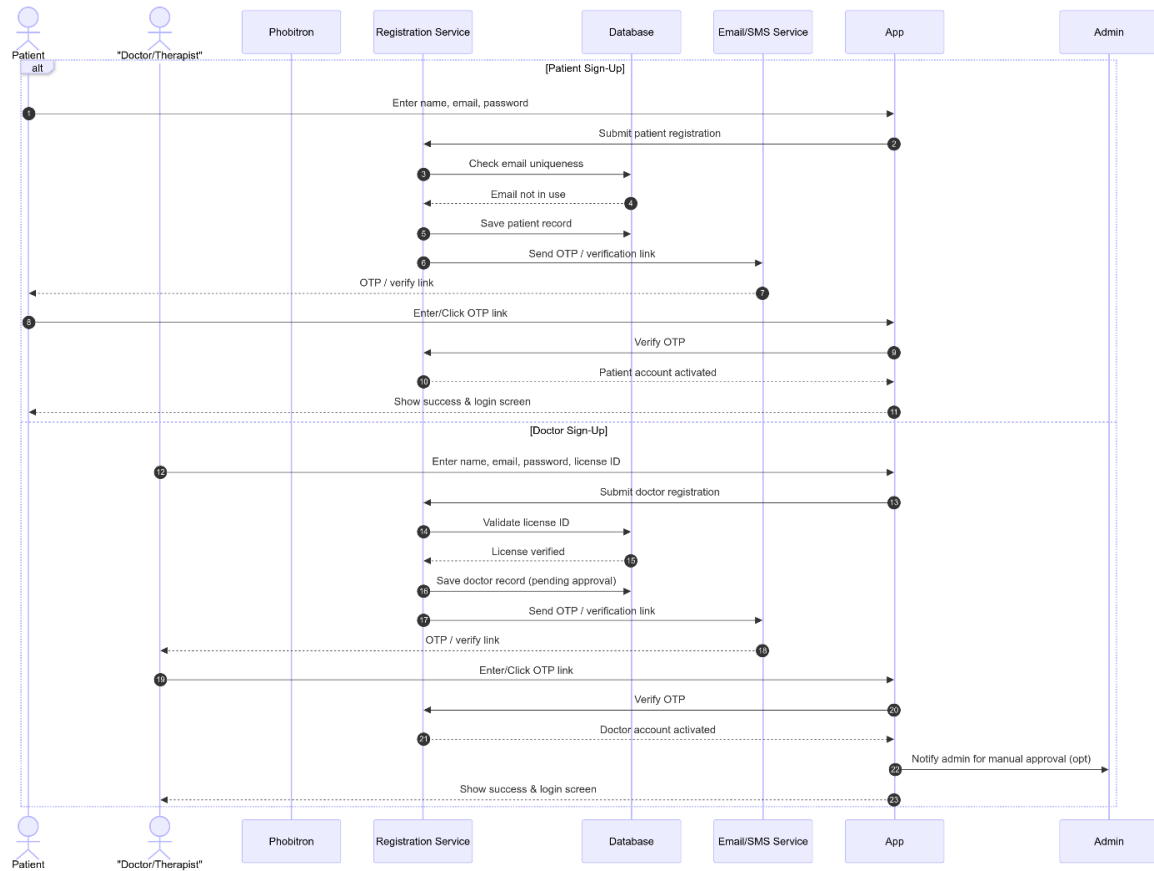


Figure 7.5: Sign-up for Patients and Medical Professional

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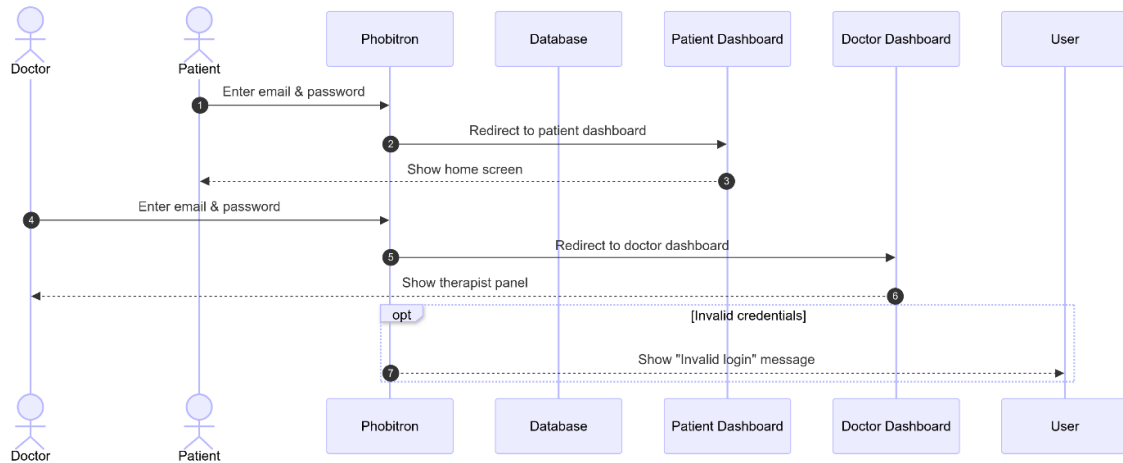


Figure 7.6: Sign-in for Patients and Medical Professionals

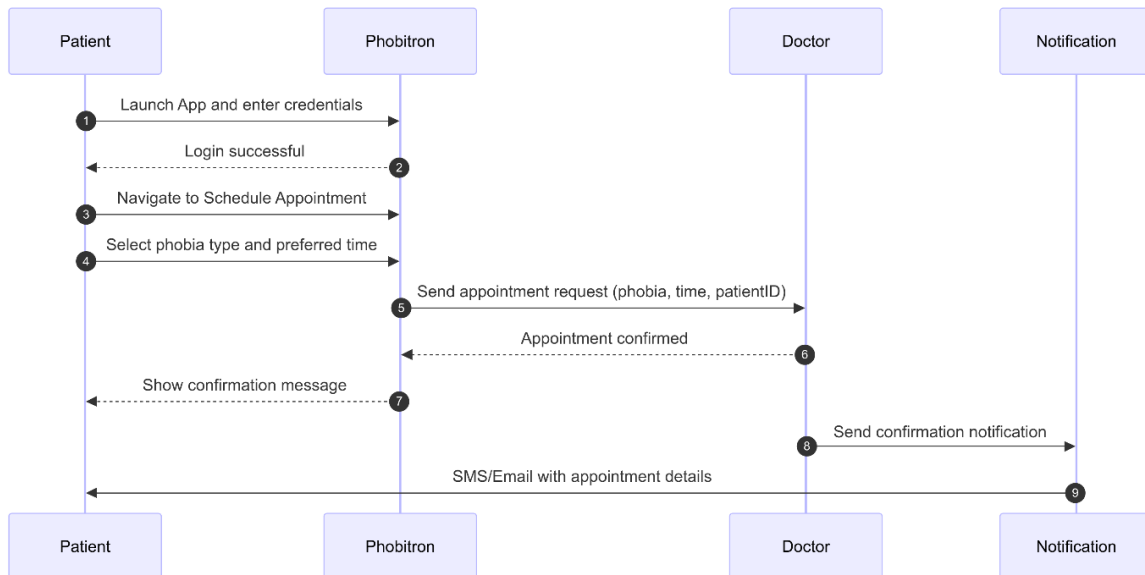


Figure 7.7: Patient booking appointments

Software Requirements Specification for Phobitron VR

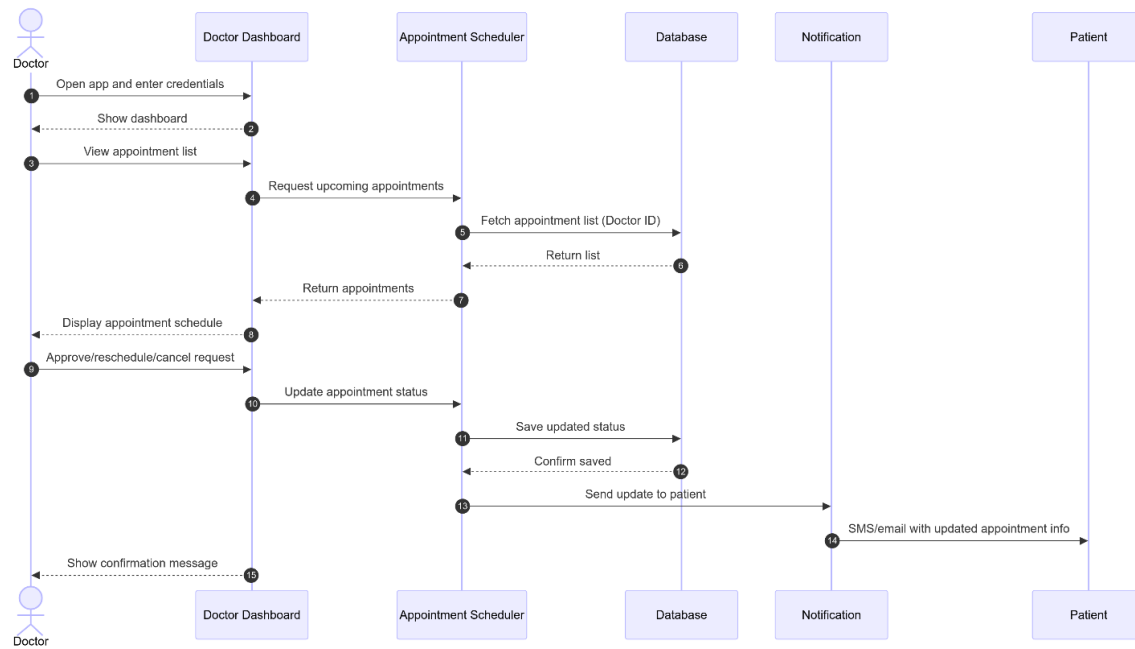


Figure 7.8: Doctor responding to appointments

Software Requirements Specification for Phobitron VR

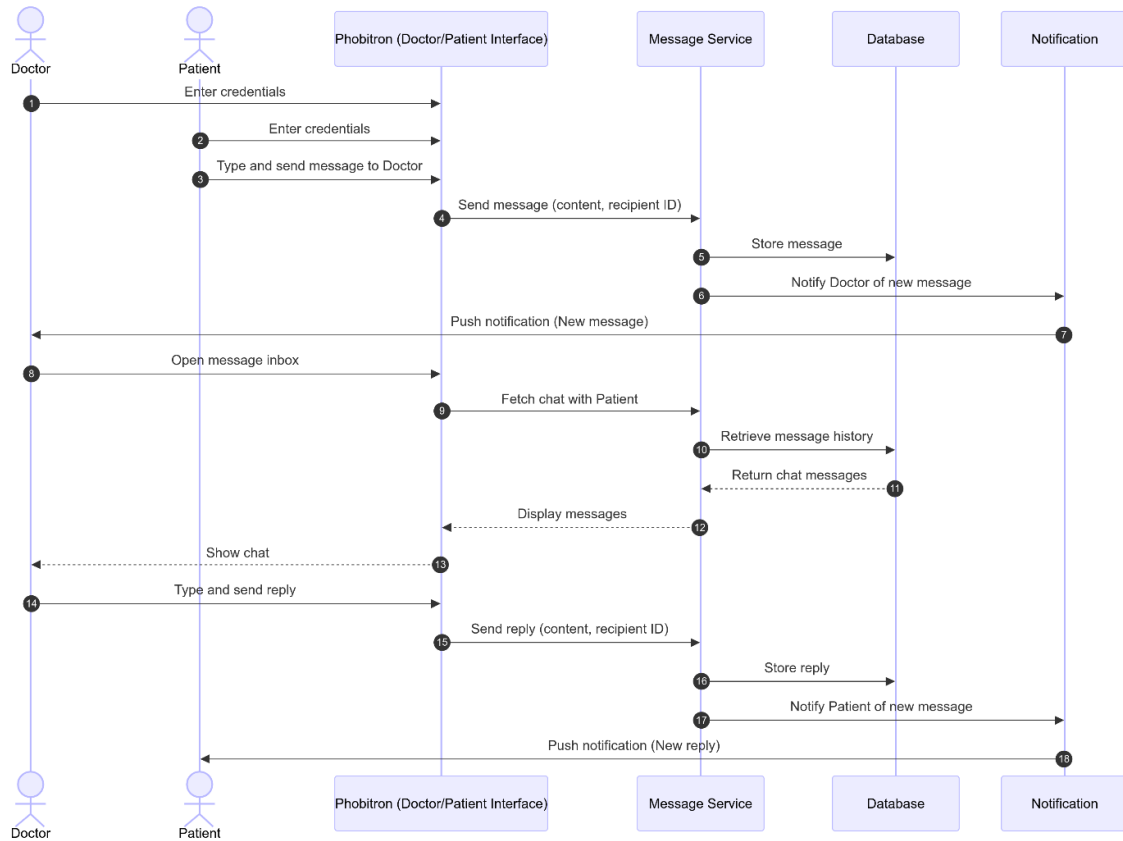


Figure 7.9: Message interaction between Patient and Doctor

Software Requirements Specification for Phobitron VR

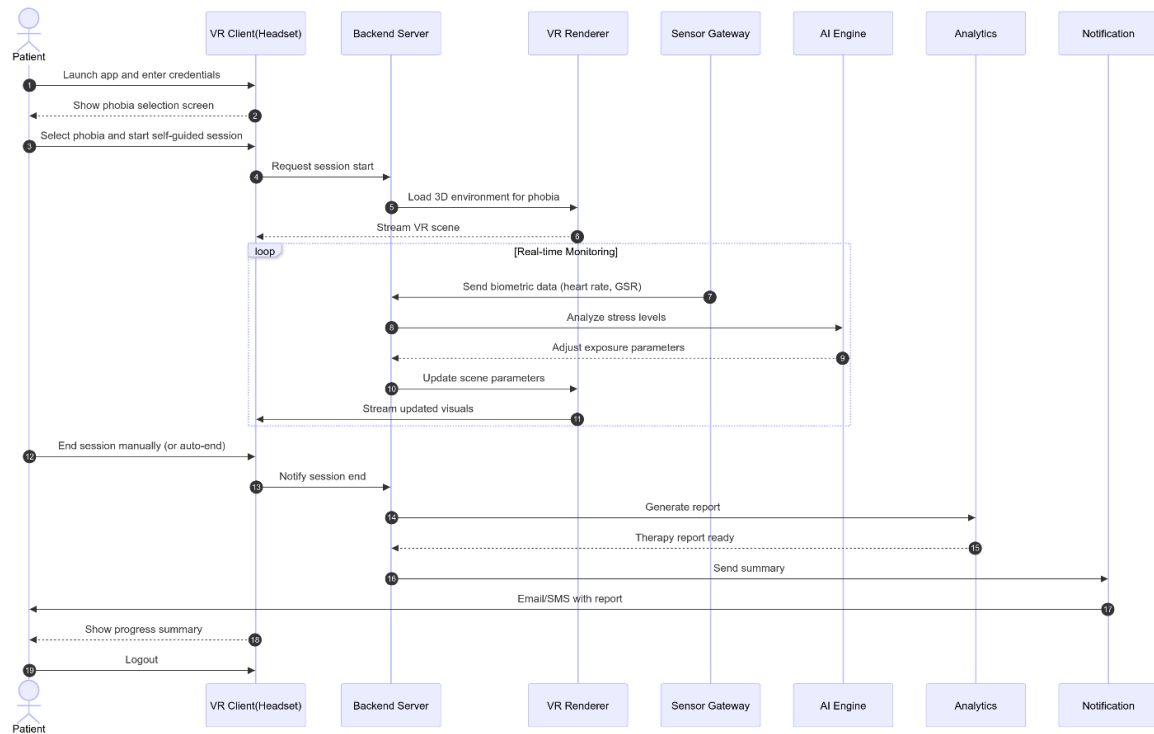


Figure 7.10: Patient's interaction with self-guided VR room

Software Requirements Specification for Phobitron VR

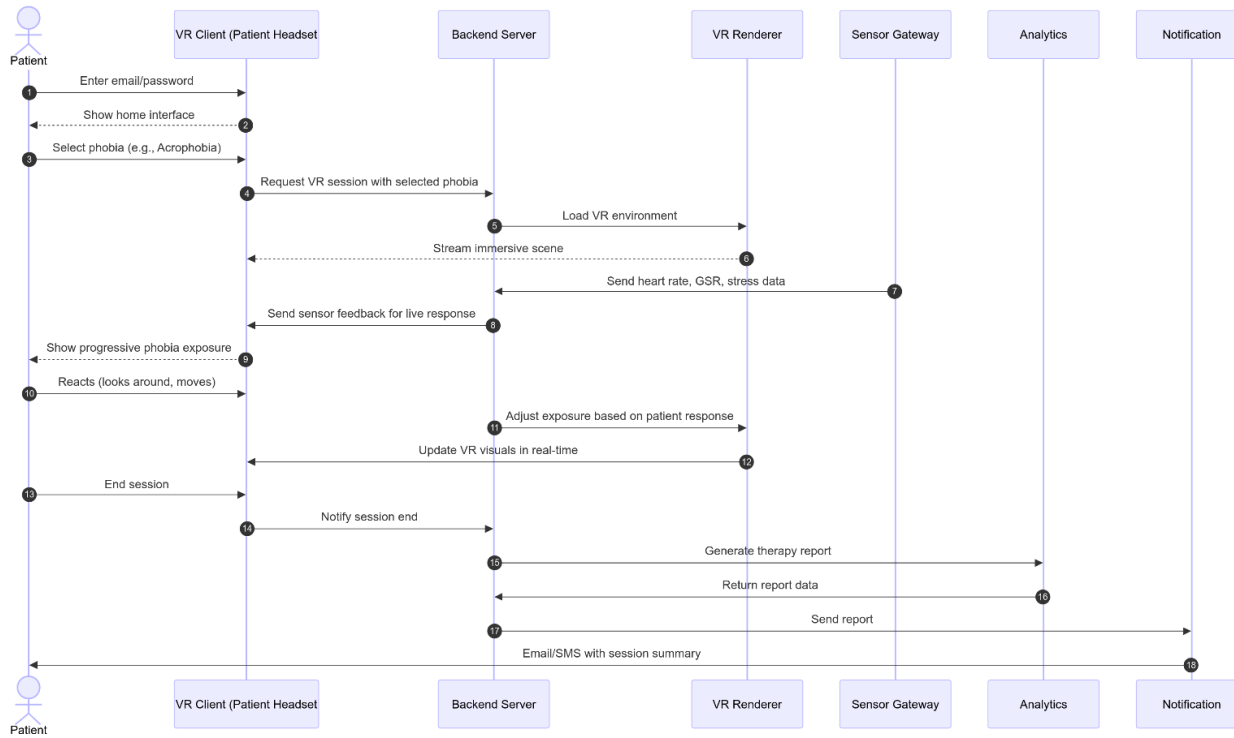


Figure 7.11: Patient's interaction with Therapy VR room

Software Requirements Specification for Phobitron VR

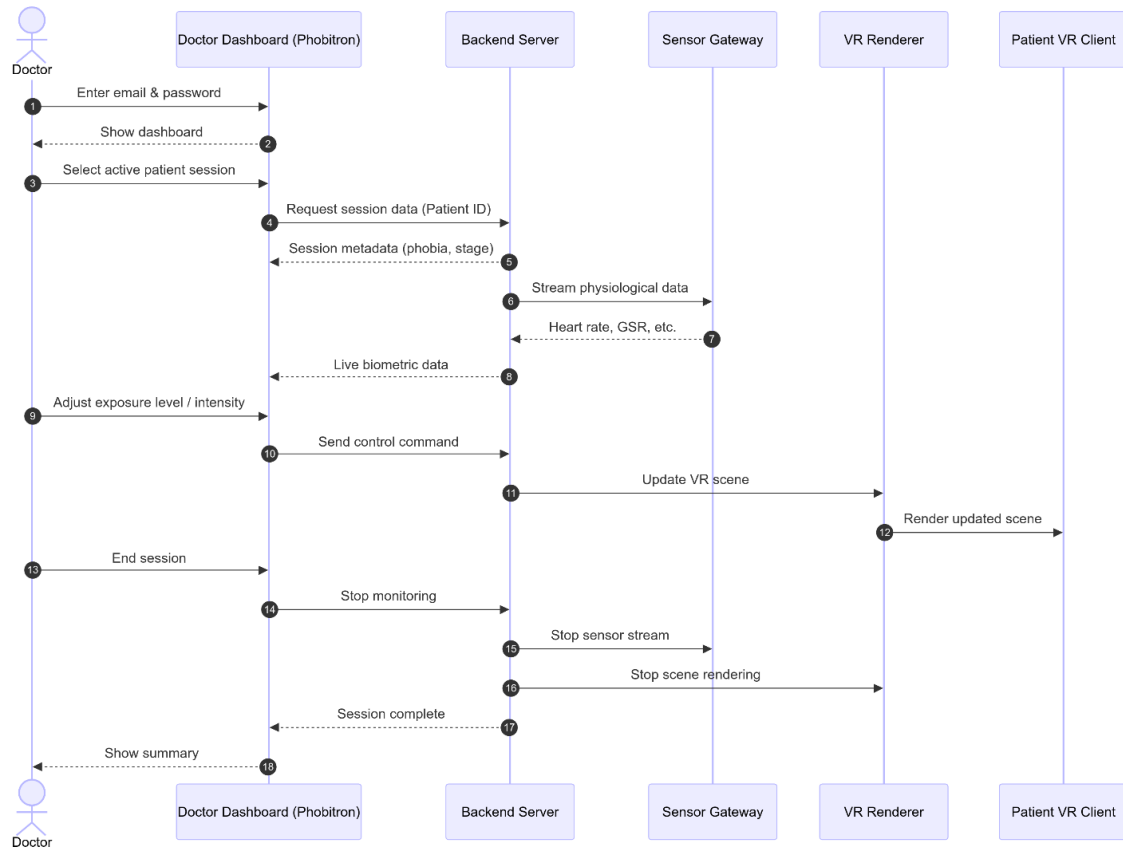


Figure 7.12: Doctor's interaction with Therapy VR room

Activity Diagram:

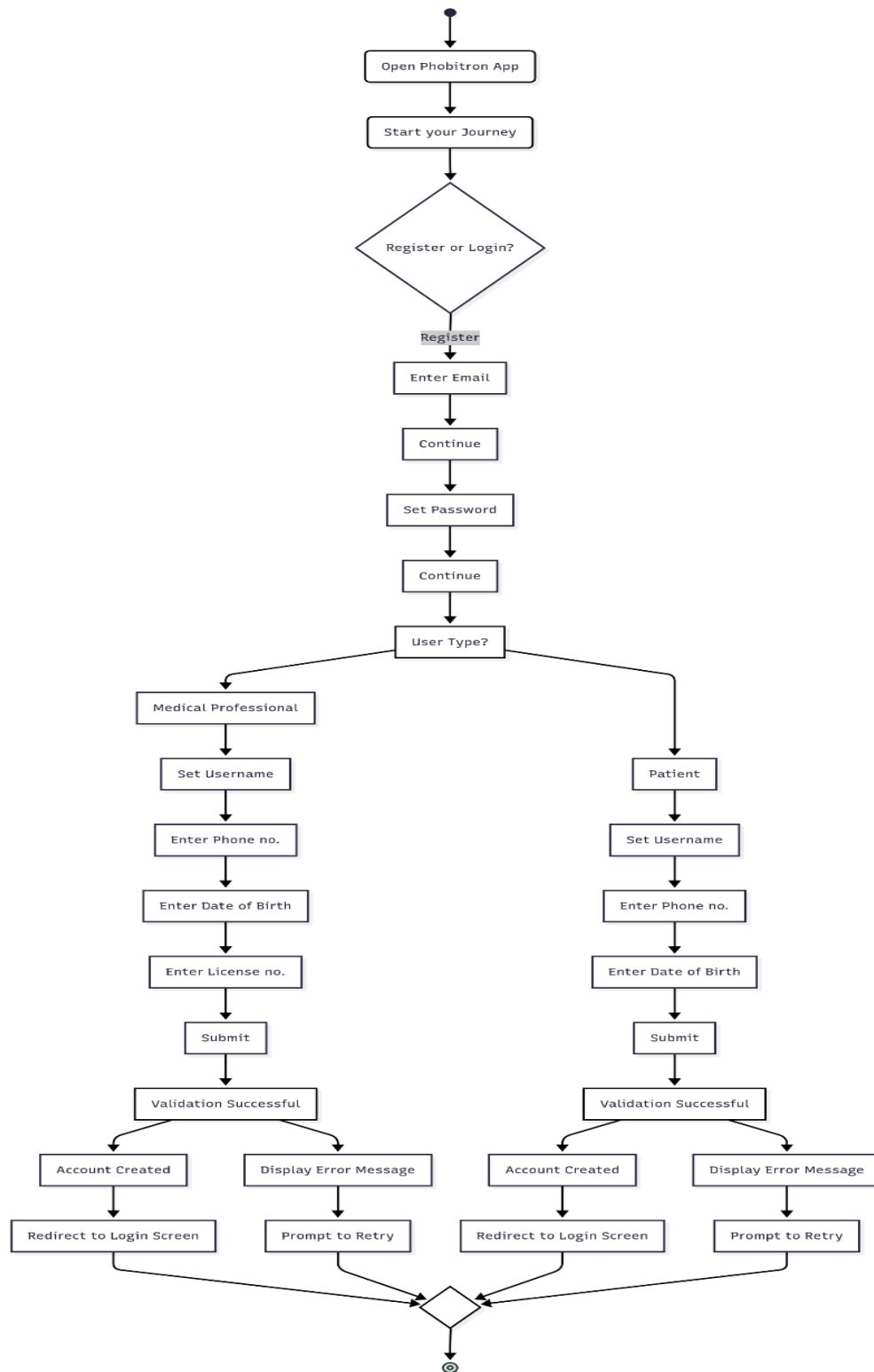


Figure 7.13: Account registration in Phobitron App

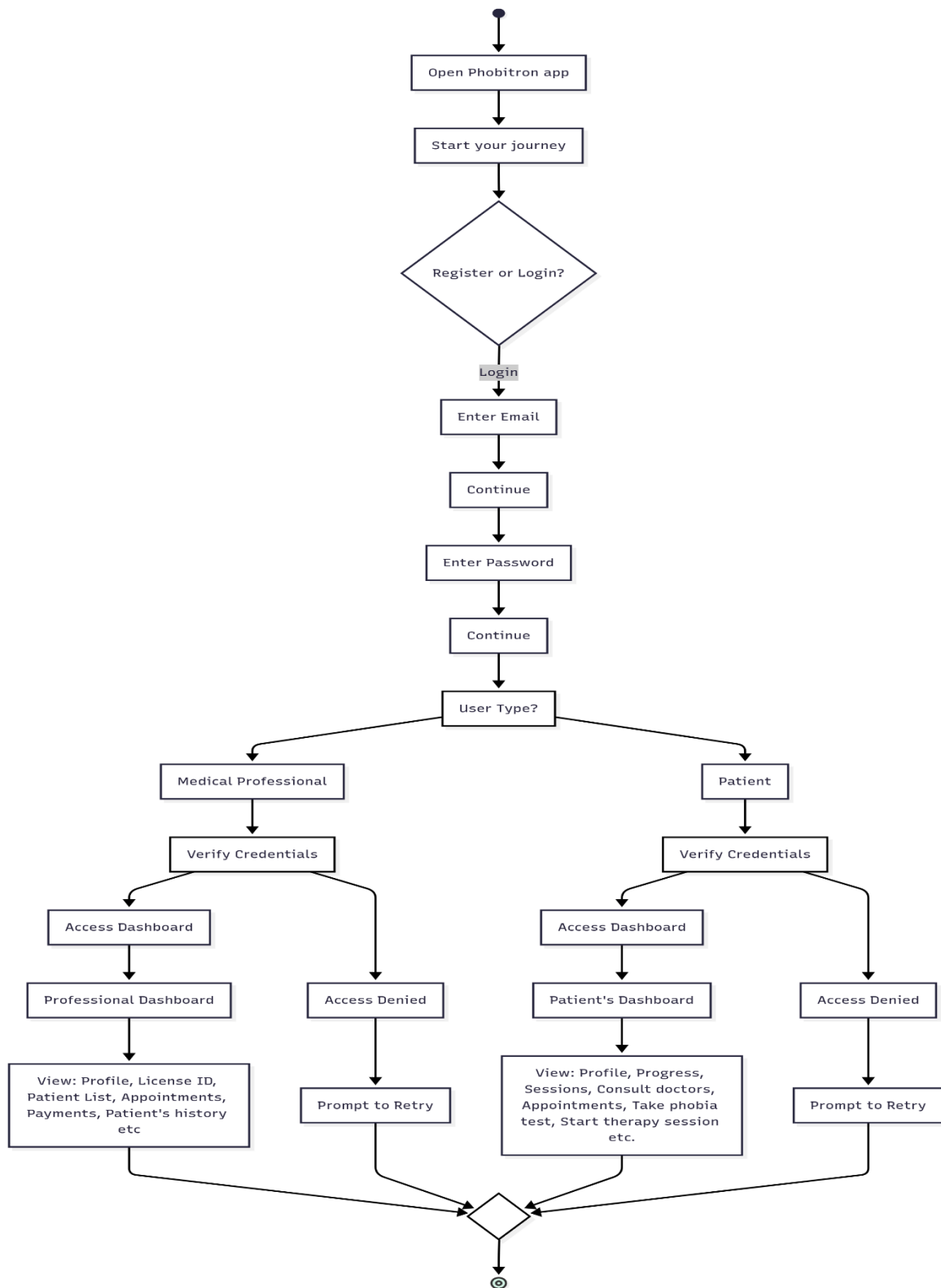


Figure 7.14: Account Log In in Phobitron App

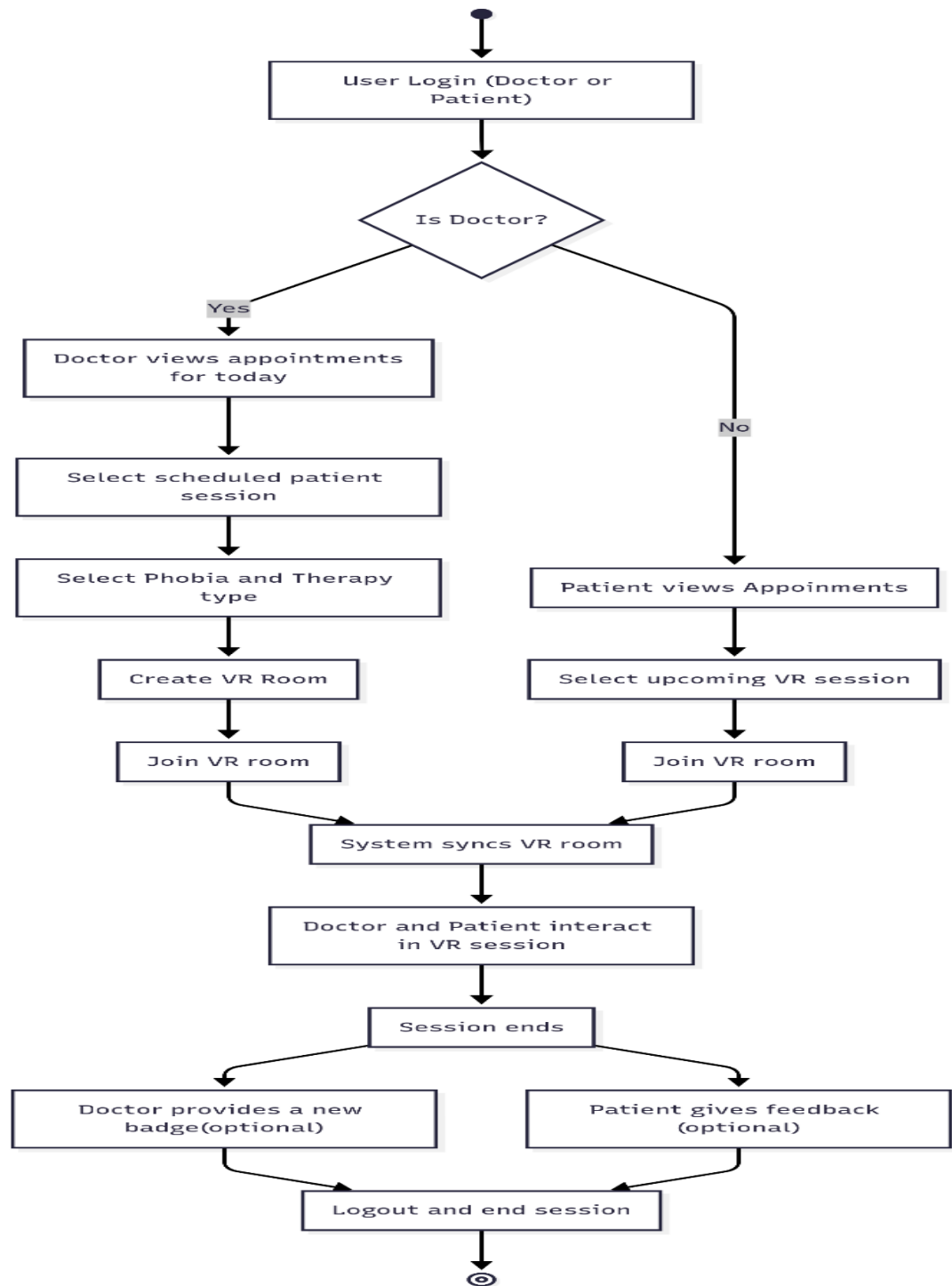


Figure 7.15: VR Guided Interaction by Doctor and Patient

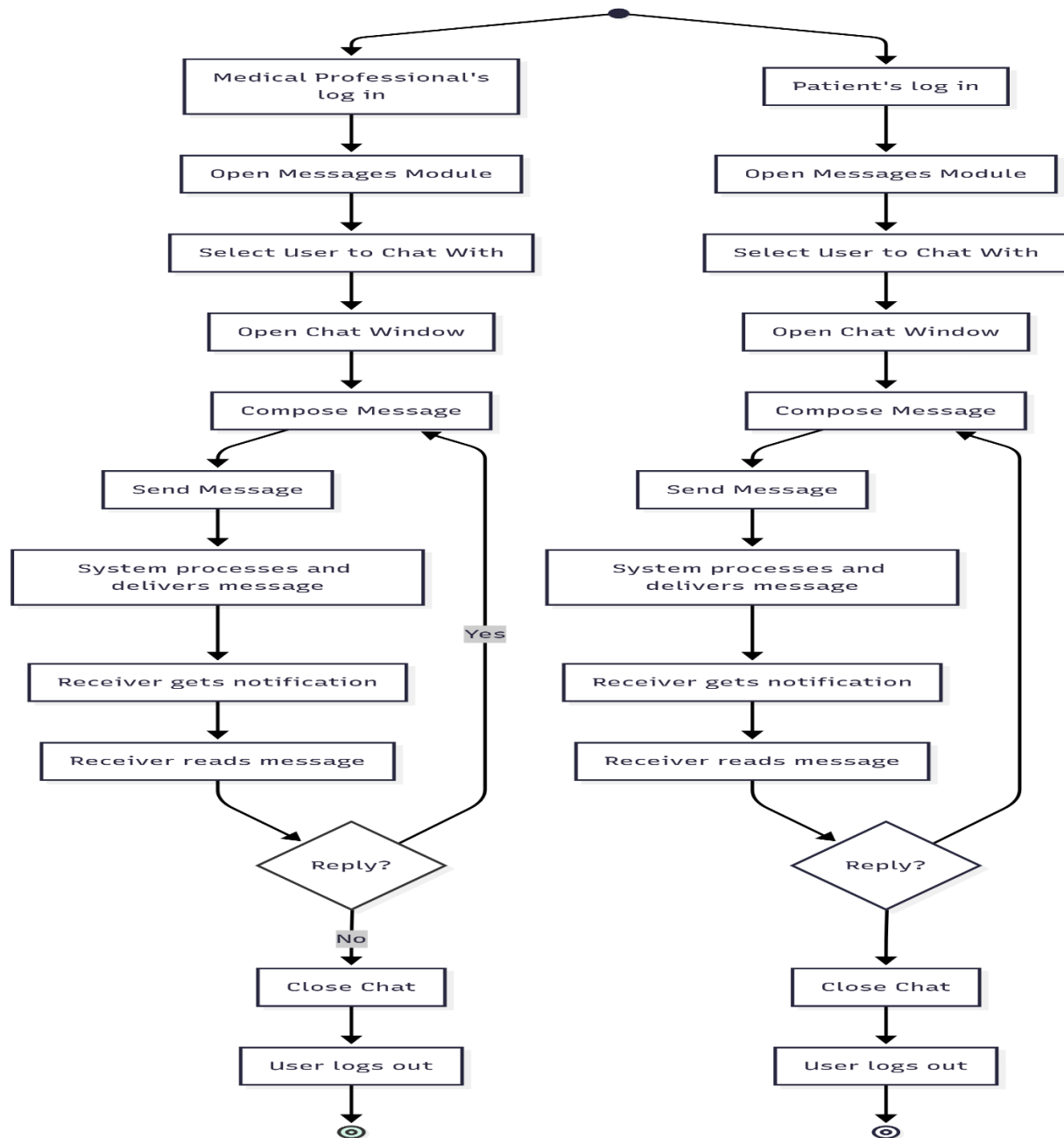


Figure 7.16: Messaging by Doctor and Patient using Phobitron

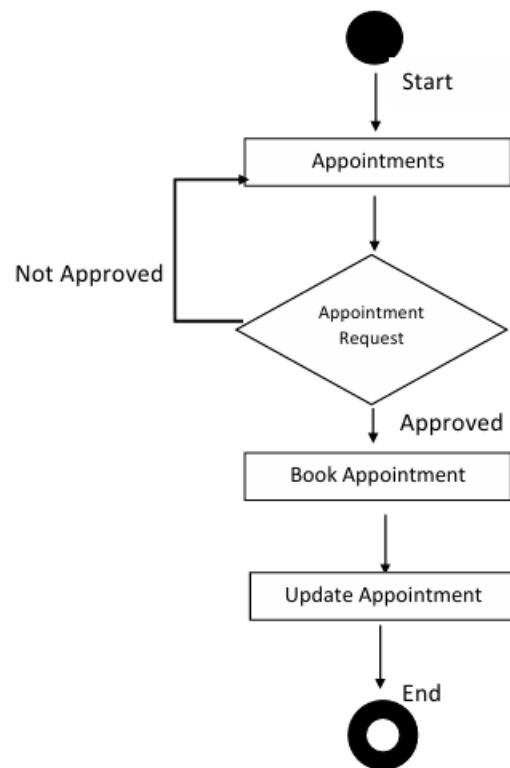


Figure 7.17: Doctor Appointment Approve Interaction in Phobitron

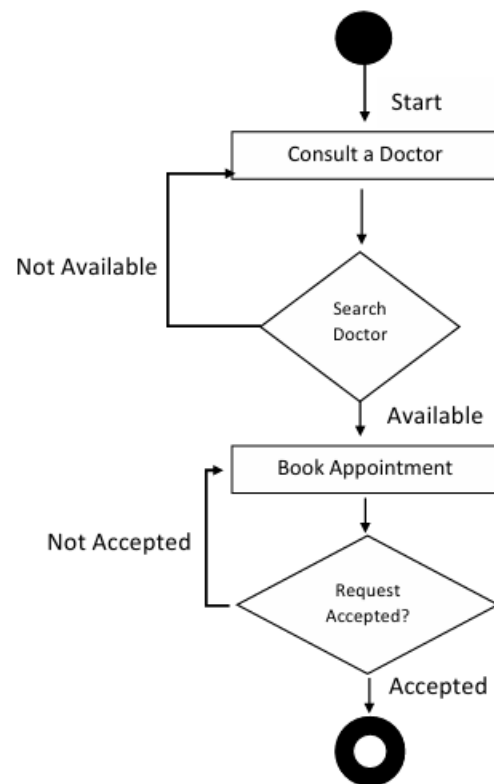


Figure 7.18: Patient's Appointment Booking Using Phobitron

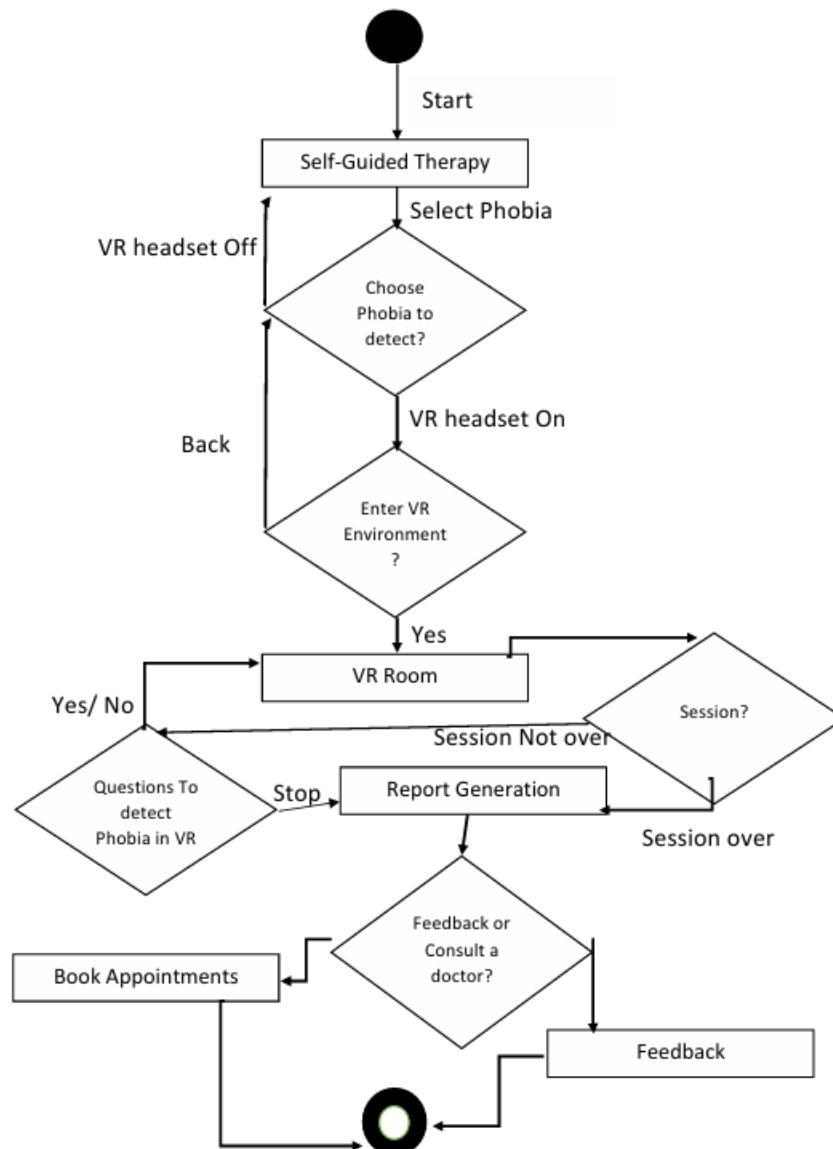


Figure 7.19: VR Self-Guided Interaction by System and Patient

Data Flow Diagram:

DFD Level 0:

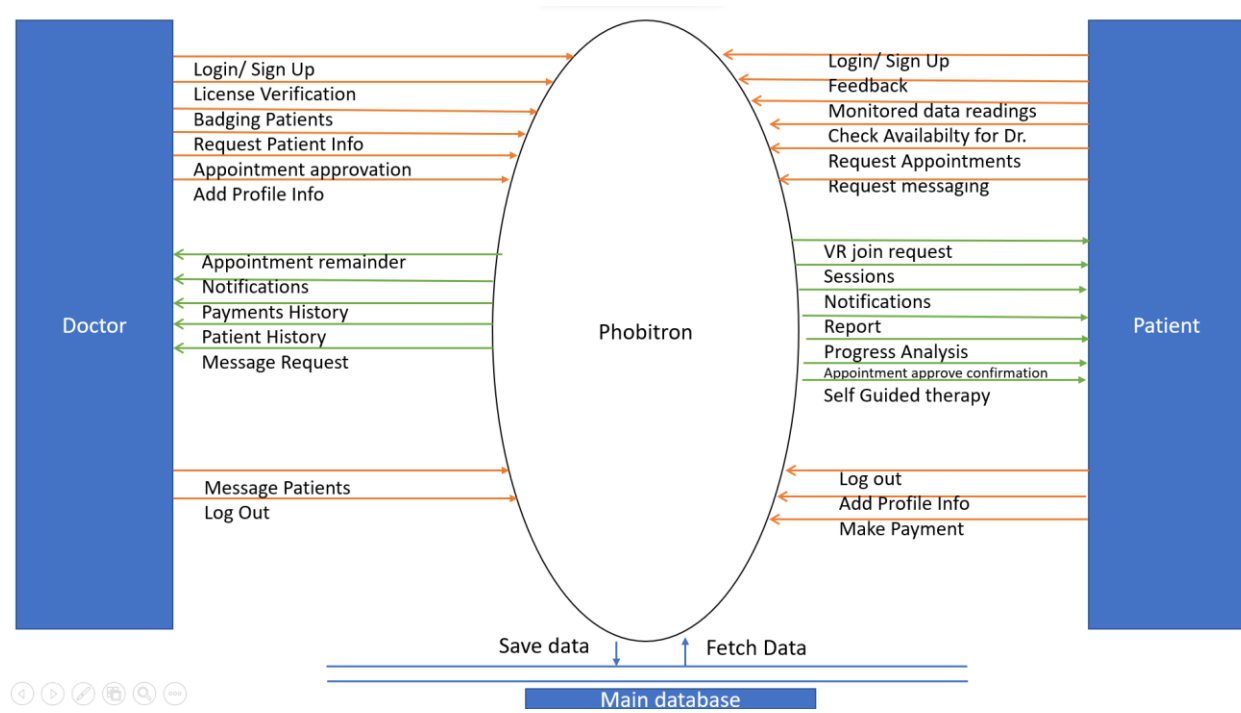


Figure 7.20: DFD 0 Level for Entire Phobitron System

Software Requirements Specification for Phobitron VR

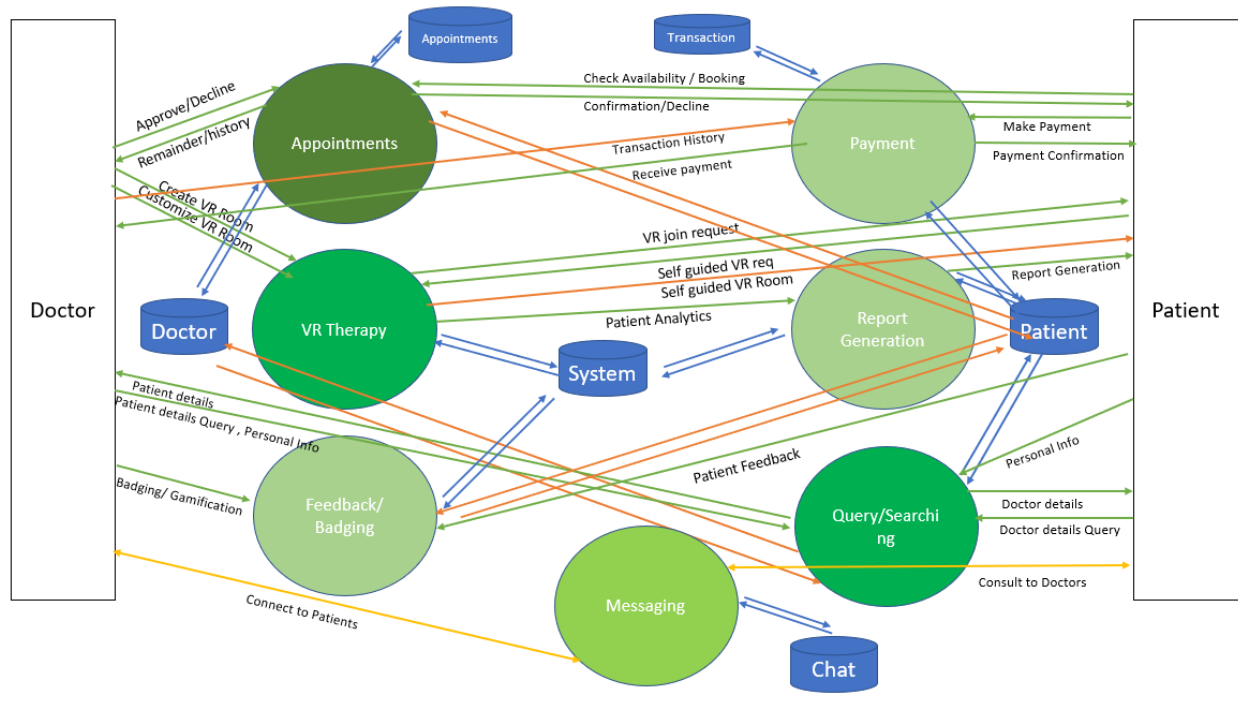


Figure 7.21: DFD 1 Level for Entire Phobitron System

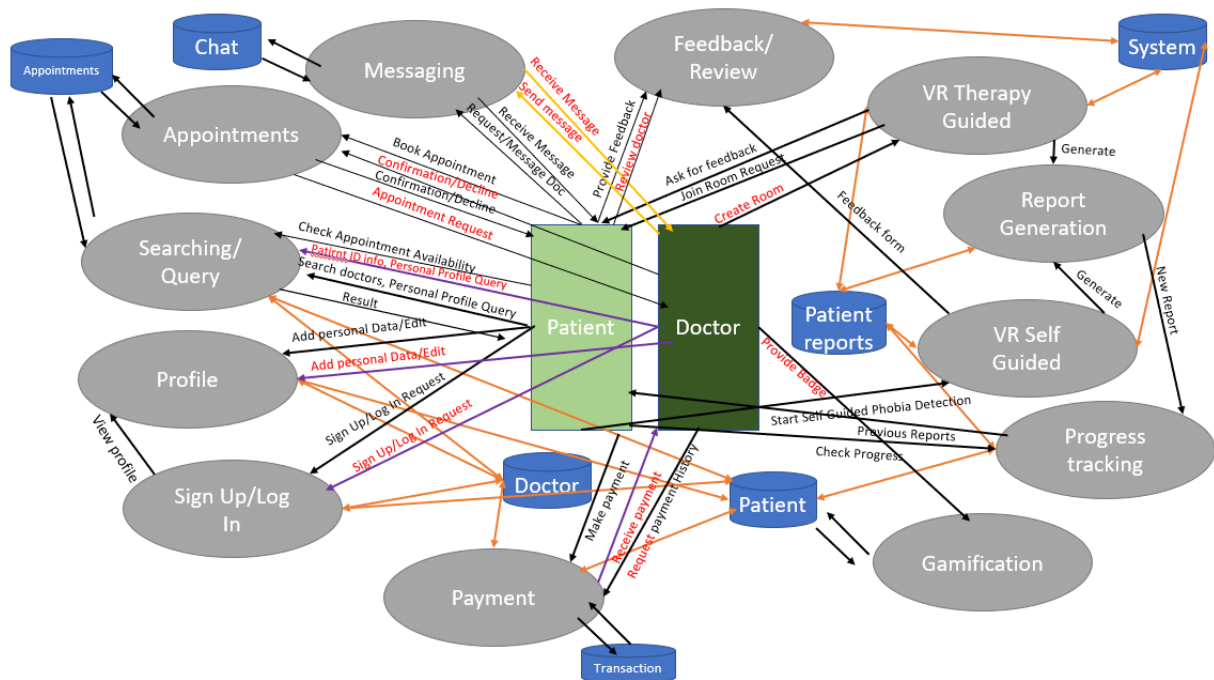


Figure 7.22: DFD 2 Level for Entire Phobitron System

8. Prototyping

8.1 Low-Fidelity Prototype

To illustrate the initial design and user interactions of **Phobitron**, low-fidelity prototypes were created. These help visualize the overall flow and interface of the VR-based phobia testing and therapy system.

The prototypes include:

- **Wireframes** of key modules such as the Sign-up Screen, Sign-in Screen, Patient's Dashboard, Doctor's Dashboard, Self-guided Phobia Test Screen, VR Therapy Session Screen, Progress Report and so on.
- Basic navigation flow showing how users move between modules.

8.1.1 Sample Mockup Screens



Figure 8.1.1.1: Phobitron Welcome Screen

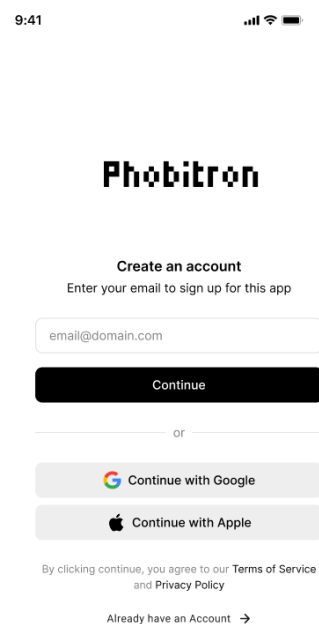


Figure 8.1.1.2: Sign-up Screen

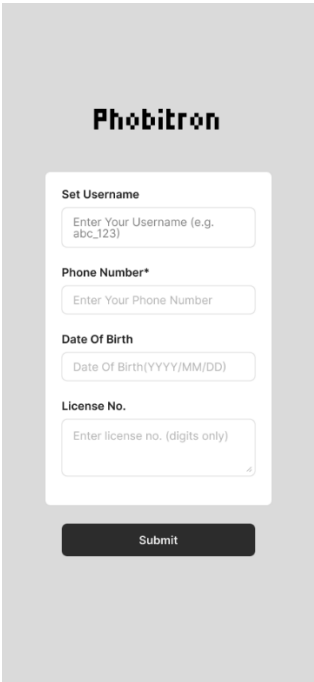


Figure 8.1.1.3:
Doctor's Sign-up Screen

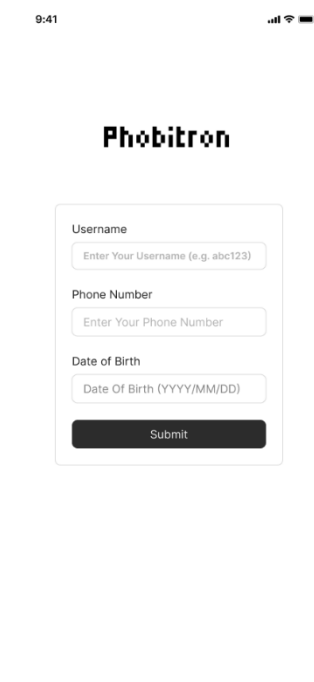


Figure 8.1.1.4: Patient's
Sign-up Screen

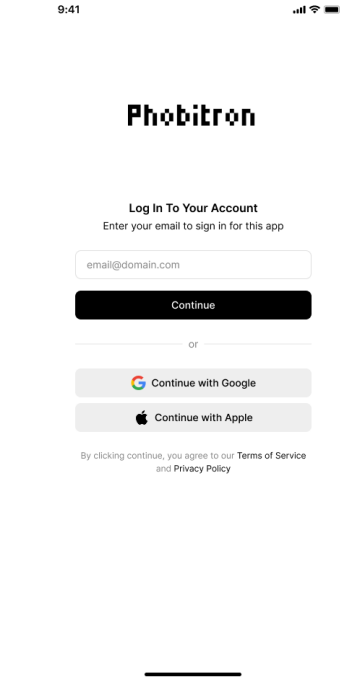


Figure 8.1.1.5: Log In
Screen

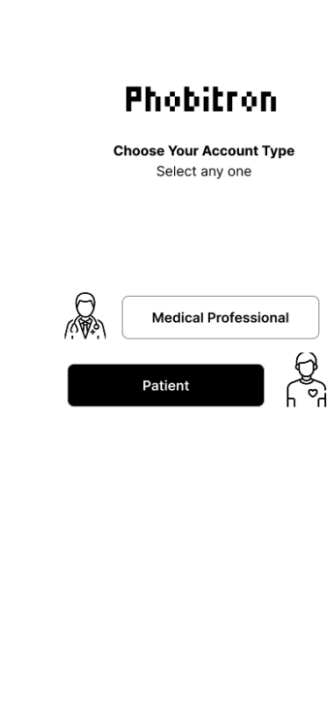


Figure 8.1.1.6:
Choosing ID Type

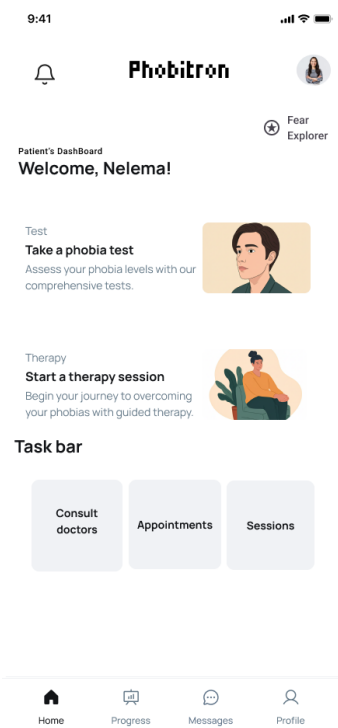


Figure 8.1.1.7:
Patient's Dashboard

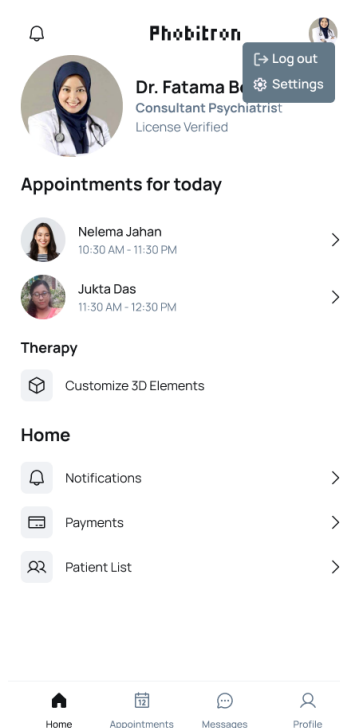


Figure 8.1.1.8:
Doctor's Dashboard

Software Requirements Specification for Phobitron VR

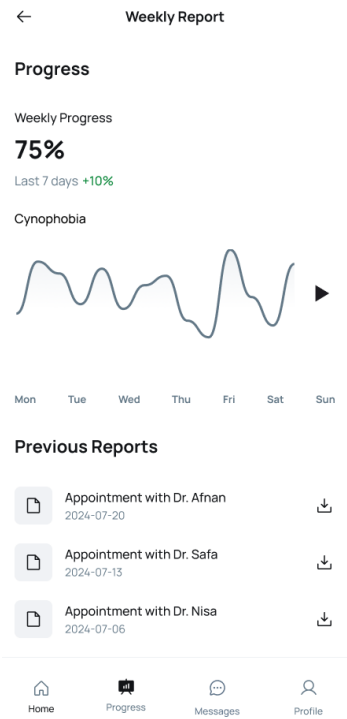


Figure 8.1.1.9: Progress Report Screen

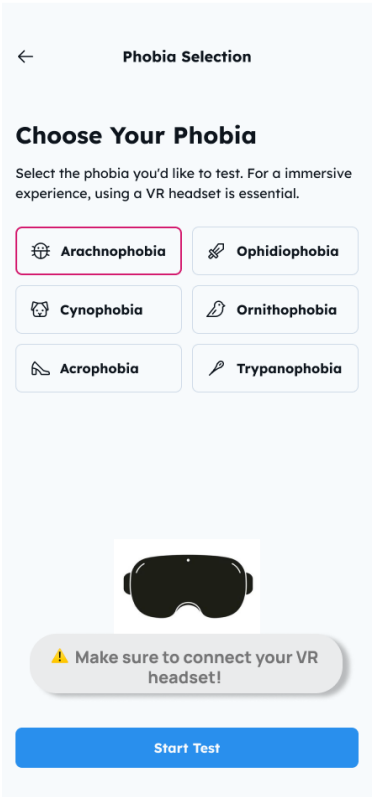


Figure 8.1.1.10: Self-guided Test Initialization Screen



Figure 8.1.1.11: Self-guided Phobia Test VR Room Screen



Figure 8.1.1.11: Doctor-guided Phobia Test VR Room Screen

8.2 Tools Used

The following tool was used to develop the prototypes:

- **Figma:** Used to design and organize wireframes, create clickable prototypes, and collaborate on interface design.

9. Advanced Requirement Management

9.1 Change Management:

To ensure that the requirements of **Phobitron** remain up to date and aligned with stakeholder needs, a formal change management process will be followed:

- **Change Request Submission:**
Team members, users, or stakeholders can submit a formal change request describing the proposed modification, its rationale, and potential impact.
- **Impact Analysis:**
The development team analyzes the change for technical feasibility, time, cost, and effect on existing requirements.
- **Approval:**
A change review board (project manager, lead developer, and product owner) reviews the impact analysis and decides whether to approve or reject the change.
- **Documentation Update:**
Approved changes are documented in the SRS and project backlog. Relevant diagrams, prototypes, and test cases are updated accordingly.
- **Communication:**
All stakeholders are notified about significant changes to keep everyone aligned.

This process helps keep requirements clear, traceable, and controlled throughout the project lifecycle.

9.2 Risk Management:

The following risks have been identified for the Phobitron project, along with proposed mitigation strategies:

Risk	Impact	Mitigation Strategy
VR headset compatibility issues	High	Test early on target devices; design modular VR scenes adaptable to different resolutions and specs
Limited user acceptance (due to discomfort or fear)	High	Conduct usability testing and gradually expose users to phobia content; provide optional guidance
Data privacy concerns (user test data and therapy reports)	Medium	Use secure storage and implement data encryption; follow best practices for data protection
Requirement changes during development	Medium	Follow the defined change management process and use agile iterations
Incomplete prototype leading to misunderstandings	Low	Regularly review and validate prototypes with stakeholders

This proactive risk management approach helps reduce project risk and keeps the development process resilient.

10. Appendix

10.1 Glossary

Terms	Explanation
VR (Virtual Reality)	A computer-generated simulation of a 3D environment that users can interact with in a seemingly real way using special equipment.
Phobia	An irrational and extreme fear of a specific object, situation, or activity.
Exposure Therapy	A psychological treatment where the patient is exposed to the feared object/situation gradually to reduce anxiety.
Desensitization	The process of reducing emotional responsiveness to a feared stimulus after repeated exposure.
Simulator Sickness	A condition similar to motion sickness, caused by prolonged VR use.
User Interface (UI)	The visual layout of an application that allows users to interact with it.
User Experience (UX)	The overall experience a user has when interacting with the app, including ease of use and satisfaction
Score/ Trigger Score	A metric used to evaluate a user's level of fear or anxiety during exposure.
Cognitive Behavioral Therapy (CBT)	A form of psychotherapy that aims to change negative thought patterns.
Biofeedback	A technique that uses sensors to monitor bodily functions like heart rate during VR sessions.
Environment Simulation	A digitally created space that mimics a real-life phobia-inducing situation.
Doctor's Dashboard	The interface used by mental health professionals to monitor, control, and assess patient therapy.
Progress Tracker	A tool or feature that monitors a user's improvement over time.
Custom Scenario	A user-specific VR situation tailored to a unique phobia.
Real-time Feedback	Immediate response or guidance provided based on user interaction.
VR Headset	The hardware device worn over the eyes to experience VR content.
Gaze Tracking	A feature that monitors where the user is looking within the VR environment.
Session Duration	The time span of one complete therapy session

Motion Sensor	A hardware component that detects movement within the VR environment.
Term	Explanation
Behavioral Response	The user's reaction to phobia triggers, such as head movement, gaze shift, or physical motion.
Session Summary Report	A post-session document containing metrics and therapist notes.
Progressive Exposure	A therapeutic technique of increasing difficulty in phobia scenarios over time.
System Requirement	The minimum hardware and software specs needed to run the VR Phobitron app.
User Authentication	Security process that verifies a user's identity before accessing the system.
AI-based Analysis	Use of artificial intelligence to assess user behavior and progress automatically.
Session Scheduling	Feature that allows users to book or get reminders for upcoming therapy sessions.

10.2 References:

(List all sources used in the development of this app, including academic papers, APIs, tools, or standards used)

Example:

- APA Guidelines for Virtual Therapy
- DSM-5 Diagnostic Criteria for Phobias
- IEEE VR Development Standards
- OpenVR SDK Documentation

10.3 Interview Notes:

Stakeholder Interview Summary

As part of the requirements gathering phase, structured interviews were conducted with two key stakeholders to understand expectations, usability needs, and features desired in the VR-based phobia therapy system "Phobitron VR." The insights helped define both functional and non-functional requirements of the system.

◇ **Stakeholder 1: Sidratur Nur**

Role: Medical Professional

Date: January 30, 2025

Key Insights:

- Emphasized the importance of **realistic simulations**, **patient progress tracking**, and **physiological monitoring** (heart rate, anxiety).
- Rated **phobia detection accuracy** as 5/5, indicating it's crucial for effective therapy.
- Identified core phobias that must be included: **acrophobia**, **claustrophobia**, **arachnophobia**, and **social anxiety disorder**.
- Supported inclusion of features like:
 - **Gradual exposure therapy levels**
 - **Real-time physiological monitoring**
 - **AI-driven recommendations**
 - **Gamification**
 - **Professional-controlled therapy sessions**
- Preferred **high customization** of VR scenarios and both **fully VR-based** and **hybrid therapy** models.
- Emphasized that **low latency is critical** to avoid therapy disruption.
- Supported **real-time dashboard tracking**, **remote therapy**, and **AI recommendations**.
- Recommended an **advanced interface for therapists** but simple interface for patients.
- Stressed the need for:
 - **Session reports with visuals and progress charts**
 - **Multi-user access** with proper security
 - **Session recording with consent**
 - **Emergency stop button for anxiety spikes**
 - **Regular reporting after every session or on demand**
- Advocated for **gamification to motivate patients**, especially younger users.
- Expressed security concerns around **data encryption** and **compliance with medical data laws**.
- Supported authentication via **username/password and OTPs**.
- Recommended supporting platforms like **Oculus Quest, HTC Vive, Windows PC, mobile, and tablets**.
- Strongly endorsed **accessibility features** such as **voice commands**, **font resizing**, **subtitles**, and **colorblind support**.
- Preferred **voice-guided instructions** for better user experience.

◇ **Stakeholder 2: Jukta Das**

Role: Phobia Patient

Date: January 30, 2025

Key Insights:

- Expected a **safe, secure, and gradual exposure system** with **feedback** and **realistic visuals**.
- Rated phobia detection accuracy as 4/5.
- Preferred focus on **acrophobia** and **arachnophobia**.
- Strongly supported features like:
 - **Customizable sessions**
 - **Real-time physiological tracking**
 - **Gamification (badges, achievements)**
 - **Therapy session recommendations**
- Chose **adjustable customization** level.
- Desired **fully VR-based, hybrid, and self-guided therapy** with optional therapist control.
- Requested **smooth, low-latency experience** to reduce stress during therapy.
- Approved **physiological tracking** only if **privacy is ensured**.
- Did not prefer **AI-based recommendations**.
- Supported **remote VR therapy**, especially for home use.
- Requested a **very simple interface**, especially for anxious users.
- Preferred **summary reports** showing improvement visually.
- Supported **limited multi-user access** – only therapist should view patient data.
- Endorsed **gamification** to make therapy engaging and stress-free.
- Opposed session recordings unless **explicit consent is obtained**.
- Strongly supported an **emergency stop button**.
- Believes **VR therapy may be effective** but expressed some uncertainty.
- Favored **report generation after every session**, weekly, monthly, or as needed by the therapist.
- Preferred **automatic notifications** for therapy reminders.
- Supported **basic login with OTPs**, discouraged complex logins.
- Recommended support for **VR and mobile devices** for session tracking.
- Strongly encouraged **accessibility features** and **voice-guided support** for ease of use.

Conclusion:

Both stakeholders support a highly immersive, customizable, and secure VR therapy system that includes:

- Realistic phobic scenarios
- Custom therapy pacing
- Remote access and voice-guided UI
- Real-time physiological monitoring
- Reporting, gamification, and therapist oversight
- Strong emphasis on user privacy and consent

These interviews directly informed feature prioritization, UI/UX decisions, platform selection, and security requirements for *Phobitron VR*.